

On-load tap-changer VACUTAP® VV

Technical Data TD 203/02



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NOTE

These technical data are intended for the calculator and designer of the transformer. These type-specific data are only valid in connection with the information contained in the general section (TD 61) since this section contains important information on such subjects as potential connection, leakage inductance, current division, and so on.

Dimensional drawings and connection diagrams are subject to change without prior notice.

Drawings submitted during bidding and ordering are always binding.

Since the on-load tap-changer is delivered to the specifications of the transformer manufacturer, the transformer manufacturer is responsible for selecting the correct properties of the on-load tap-changer so that the requirements of the transformer are met.

1 General

1 General

1.1 Summary of the technical data

On-load tap-changer	VVIII 250 Y		VVIII 250 D		VVIII 400 Y		VVIII 400 D		VVIII 600 Y		VVIII 600 D		VVI 401 ³⁾		VVI 401 ³⁾		
Number of poles and application	3 neutral point		3 at any position on the winding		3 neutral point		3 at any position on the winding		3 neutral point		3 at any position on the winding		1 at any position on the winding				
Max. rated through-current I _{um} (in A)	250				400				600				400				
Rated short-time withstand current (in kA)	4				5				6				5				
Rated short-circuit duration (in s)	3				3				3				3				
Rated peak withstand current (in kA)	10				12.5				15				12.5				
Max. rated step voltage U _i (in V)	2000				2000 ... 1700 ¹⁾				2000 ... 1000 ¹⁾				2000 ... 1700 ¹				
Step capacity (in kVA)	see diagram P _{ST} -/I _u																
Rated frequency (in Hz)	50 ... 60																
Operating positions	without change-over selector: max. 12, with change-over selector max. 23												with change-over selector max. 23				
Rated insulation level																	
Highest voltage for equipment U _m (in kV) ²⁾	40	76	40	76	145 ⁴⁾	40	76	40	76	145 ⁴⁾	40	76	40	76	145 ⁴⁾	76	145 ⁴⁾
Rated lightning impulse withstand voltage (in kV, 1.2 I 50)	200	350	200	350	650	200	350	200	350	650	200	350	200	350	650	350	650
AC withstand voltage (in kV, 50 Hz, 1 min.)	70	140	70	140	275	70	140	70	140	275	70	140	70	140	275	140	275
Rated withstand voltages of internal insulation	see table 3																
Oil compartment	Pressure-proof up to 0.3 bar, continuous difference pressure (test pressure 0.6 bar), head and cover of the on-load tap-changer are vacuum-proof.																
Temperature range	The on-load tap-changer VACUTAP® VV can be operated in the rated load range with oil temperatures of -25 °C to +105 °C.																

Table 1a



On-load tap-changer	VV III 40 kV		VV III 76 kV		VV III 145 kV		VV I 76 kV		VV I 145 kV	
Weight without oil (ca. in kg)	240		255		280		200		215	
Displacement volume (in dm ³)	295		328		405		148		178	
Oil filling quantity V_s and minimum volume ΔV ⁵⁾ of the diverter switch oil compartment (in dm ³)	V_s	ΔV	V_s	ΔV	V_s	ΔV	V_s	ΔV	V_s	ΔV
	240	30	265	33	325	39	125	16	145	19

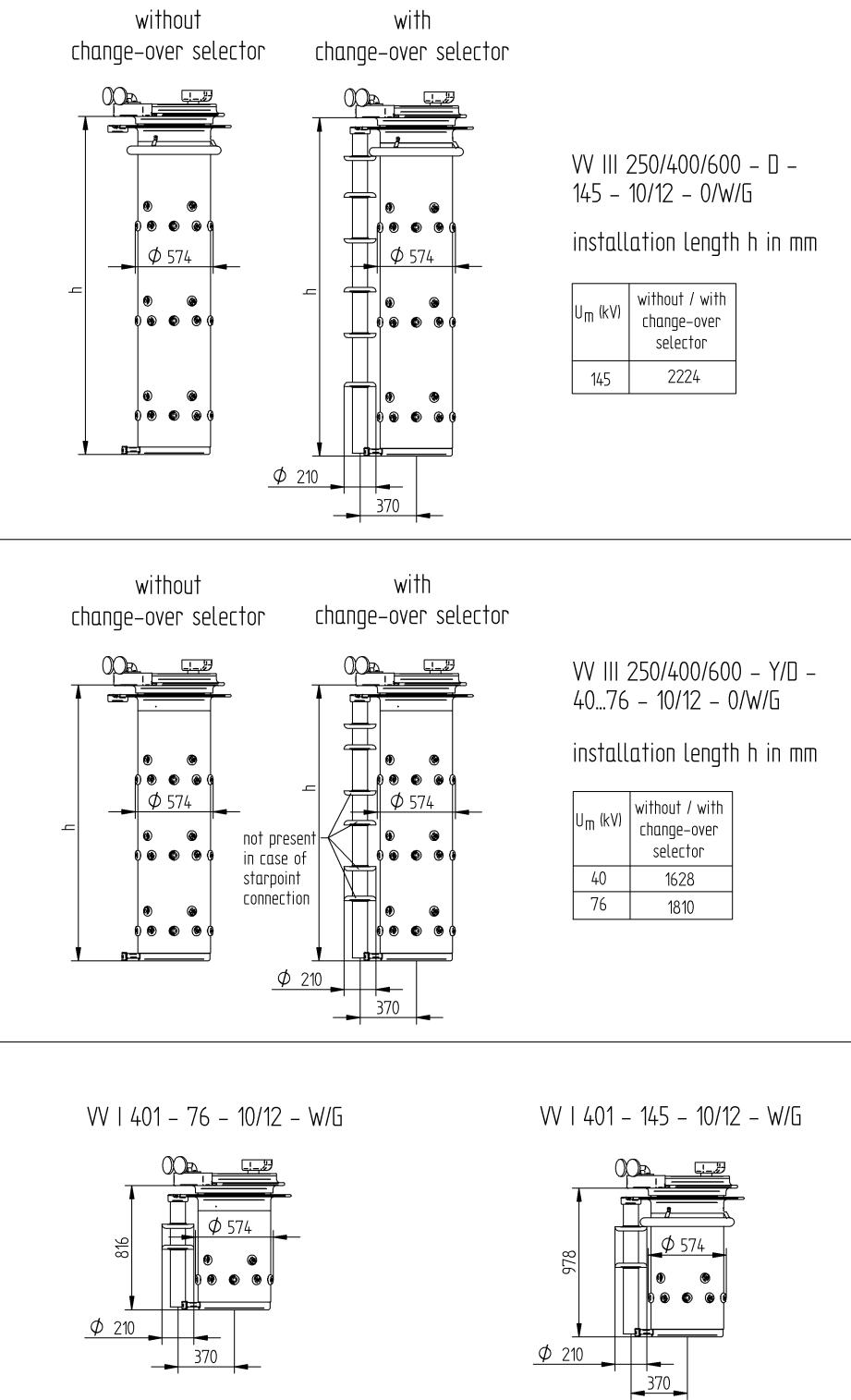
Tabelle 1b

Notes for **table 1a** and **table 1b**:

- ¹⁾ See also diagram U_i / I_u .
- ²⁾ In acc. w. VDE 0111, part 1: r.m.s. value of the phase-phase voltage for which a piece of equipment is rated for its insulation
- ³⁾ 600 A model on request.
- ⁴⁾ A max. operating voltage of 132 kV + 15 % = 151.8 kV is admissible if the test voltages for the 145 kV series are not exceeded.
- ⁵⁾ Applies for oil temperature $\vartheta = -30\text{ °C} \dots +100\text{ °C}$.

1 General

1.2 Survey

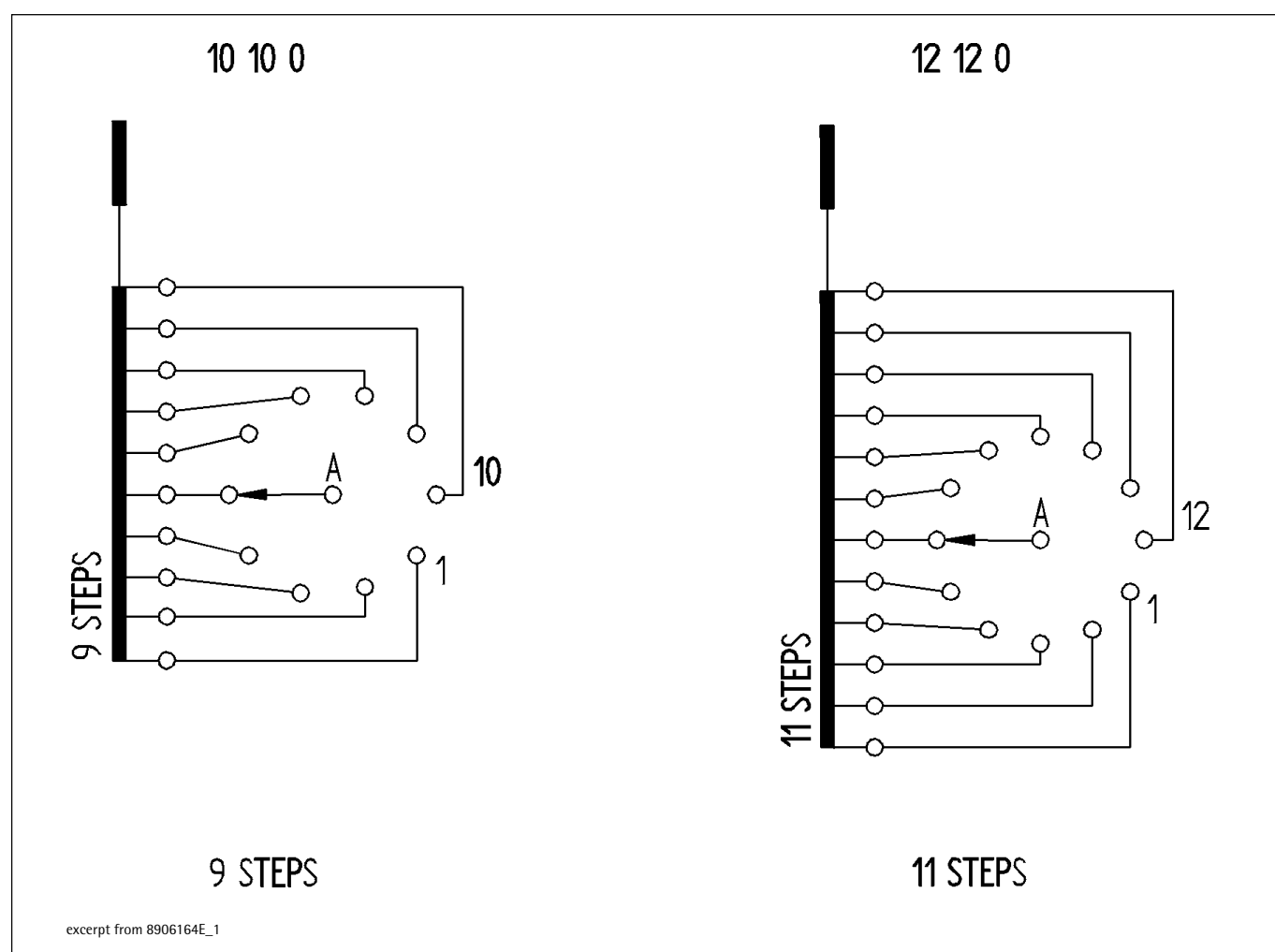


excerpt from 8995411E

Fig. 1 Survey of the designs (no. of poles, change-over selector and installation length)

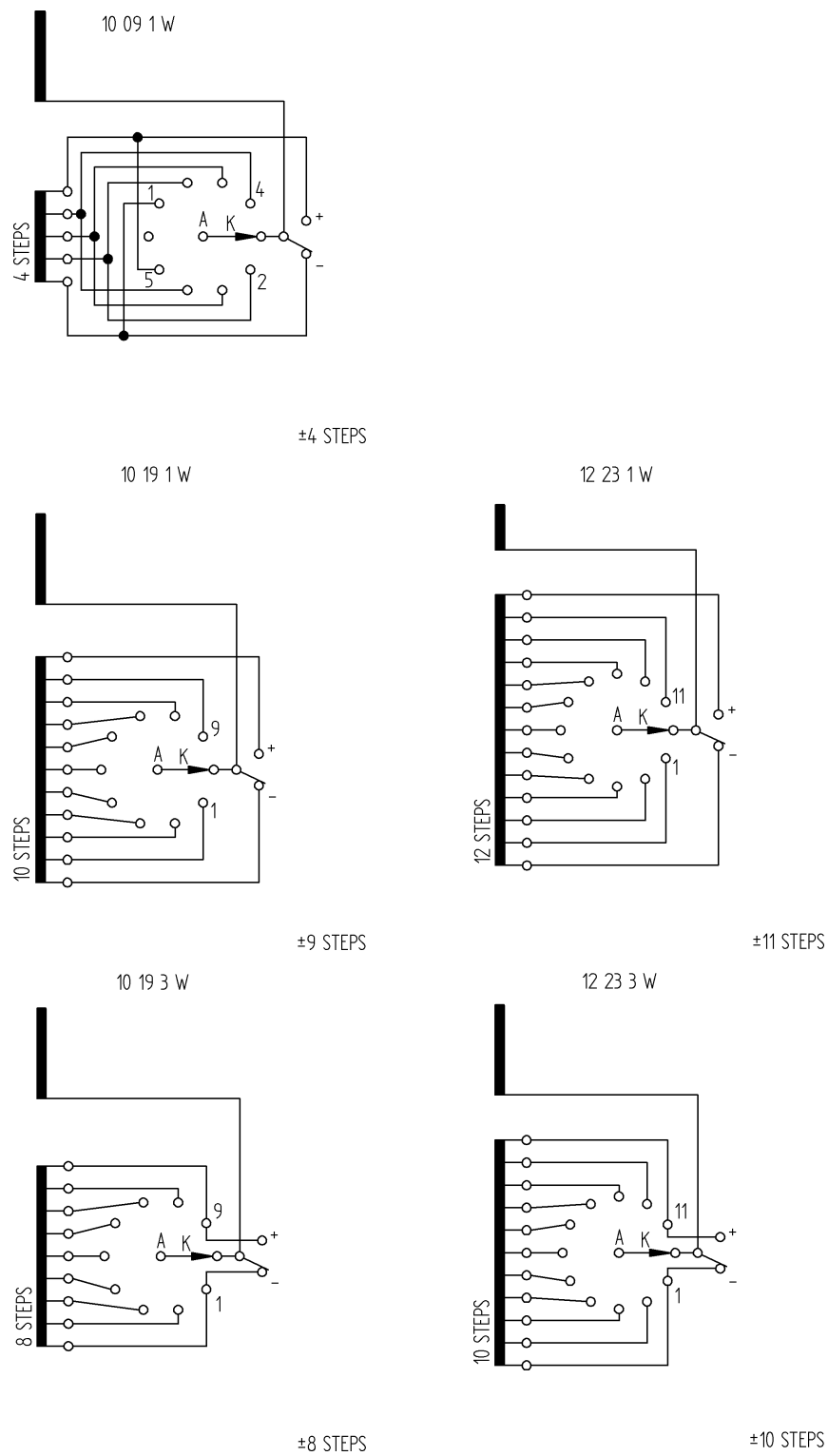
Survey of the basic connection diagrams (**fig. 2a, 2b, 2c**) with designation of the tap selector terminals in accordance with MR standards. This contact designation corresponds to the specifications in the on-load tap-changer dimension drawings.

Fig. 2a



1 General

Fig. 2b



excerpt from 8906164E_2

Fig. 2c

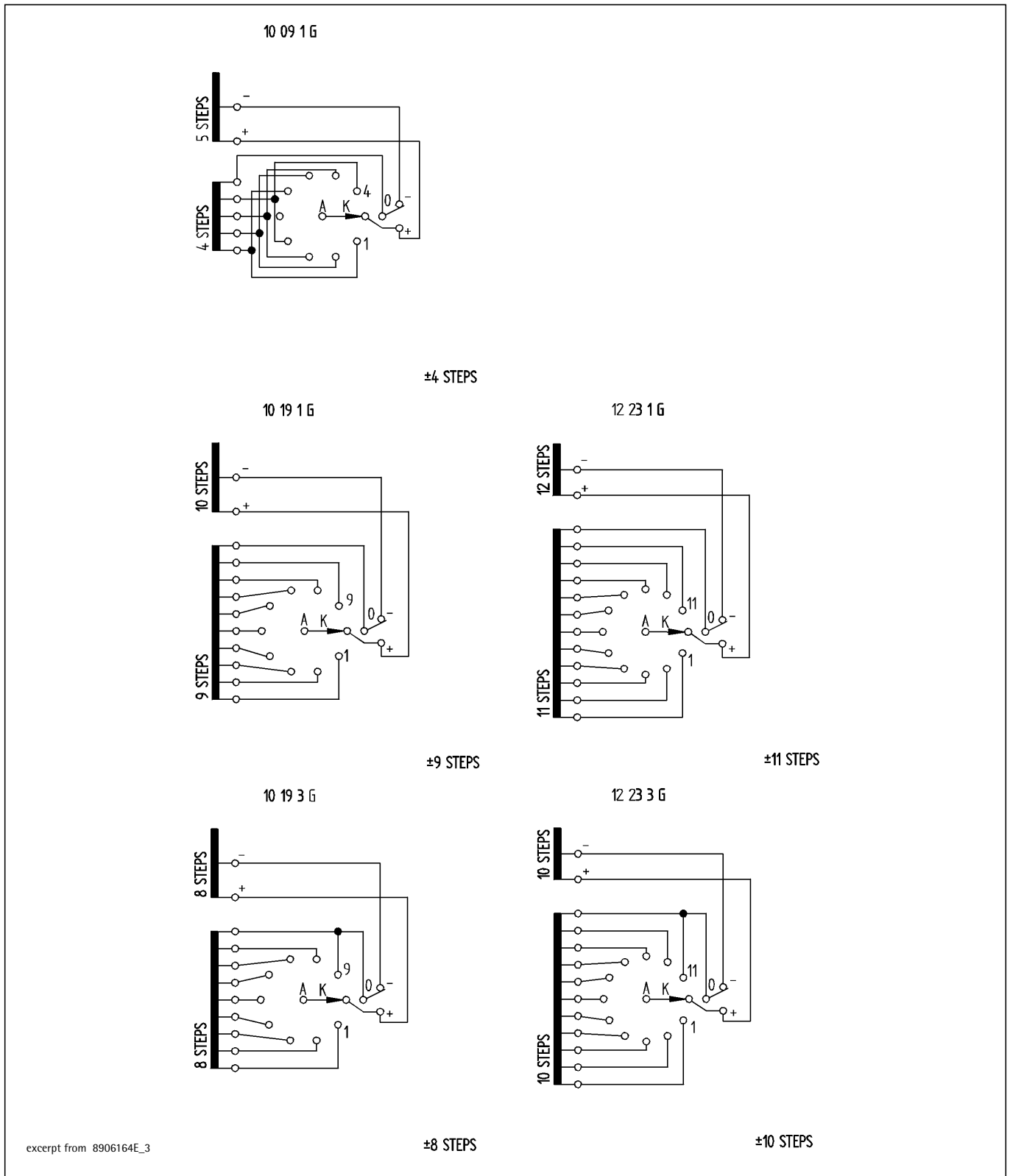


Fig. 3 Example: On-load tap-changer **VV III Y**, basic connection diagram 10 10 0

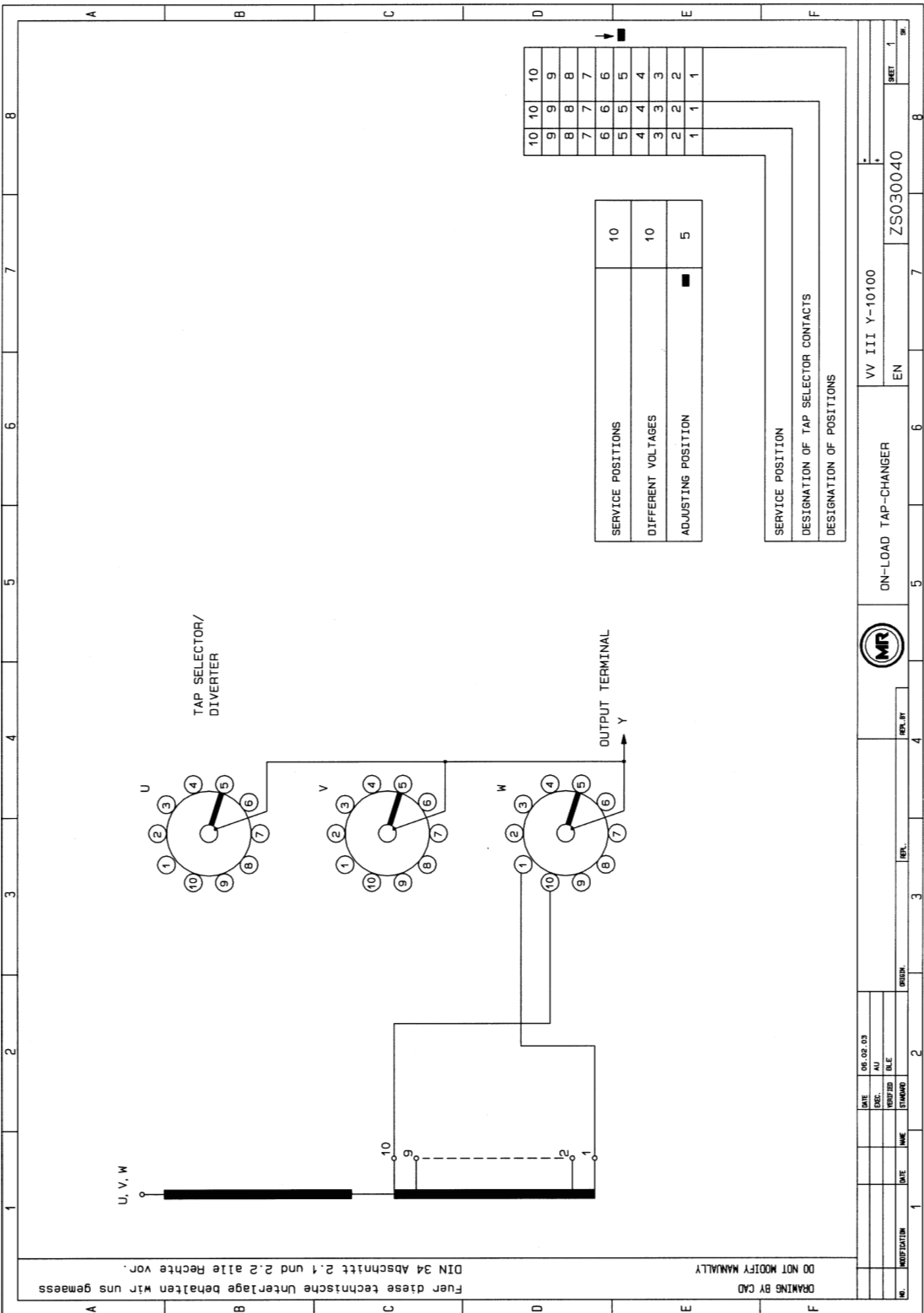


Fig. 4 Example: On-load tap-changer **VV III Y**, basic connection diagram 10 19 1 W

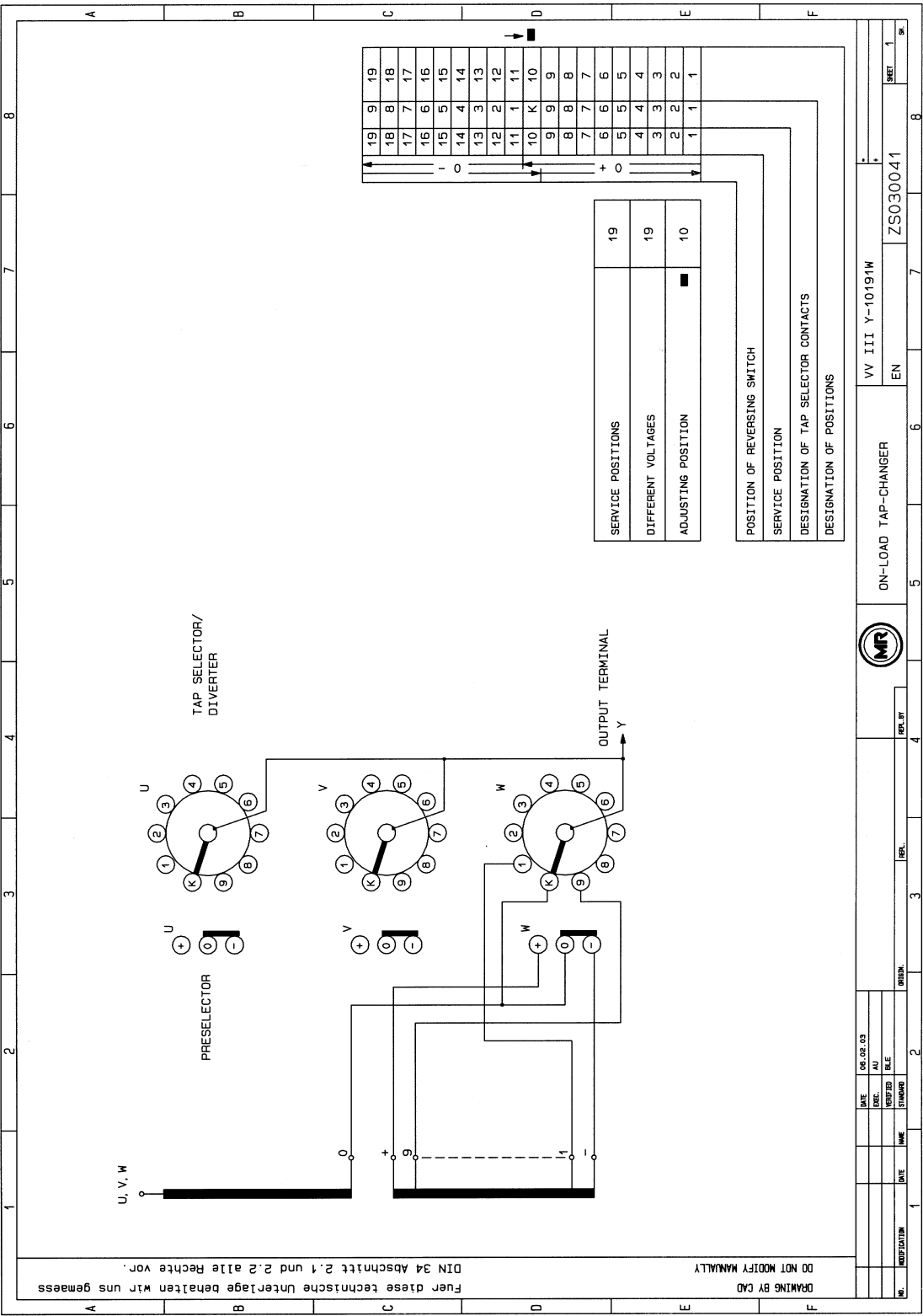
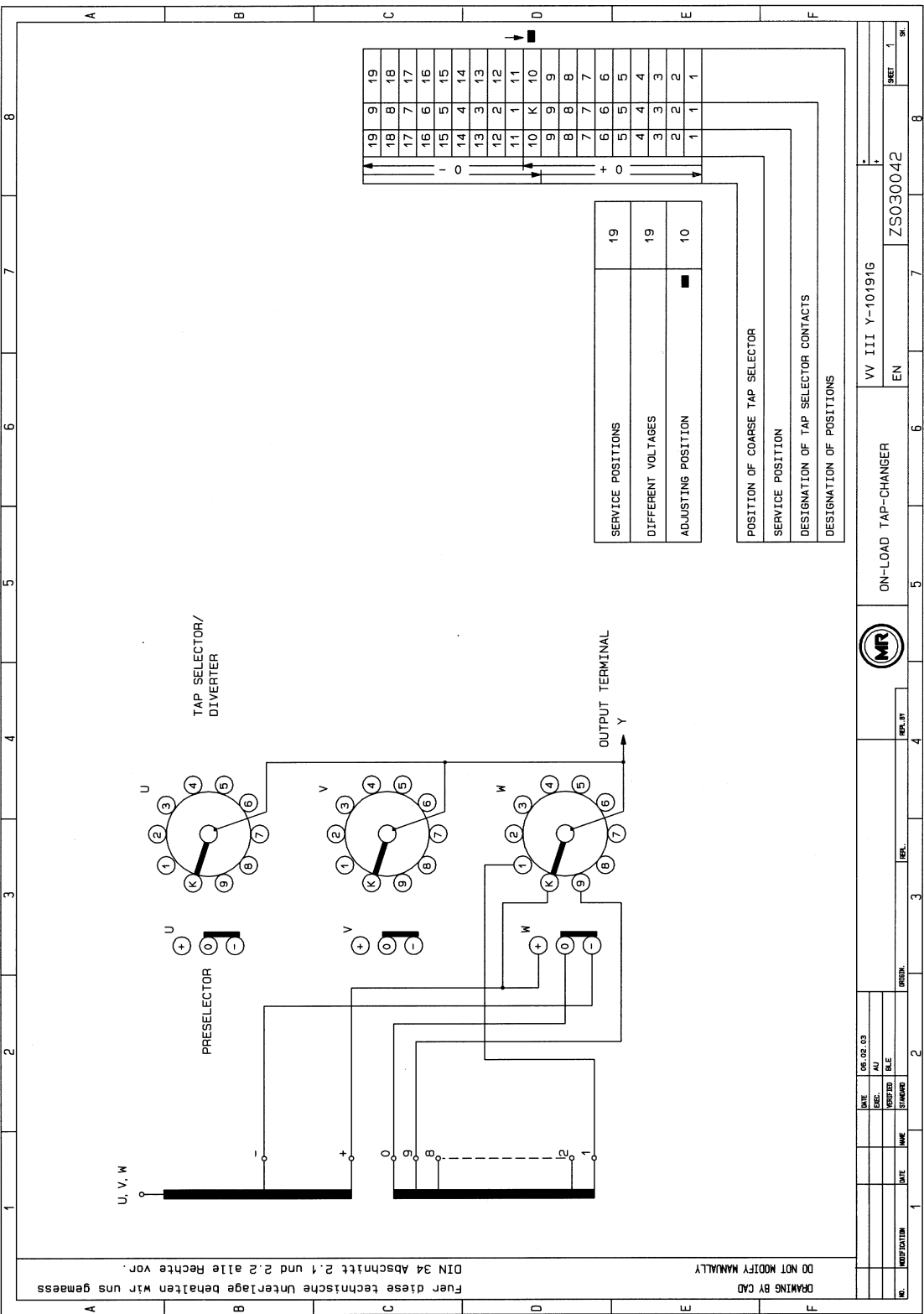


Fig. 5 Example: On-load tap-changer **VV III Y**, basic connection diagram 10 19 1 G



2 Technical Data

2.1 Rated through-current (I_U), rated step voltage (U_i) and step capacity (P_{StN})

On-load tap-changer	VV III 250	VV III 400 / VV I 401	VV III 600
Contacts	10/12	10/12	10/12
I_U (in A)	250	400	600
U_i (in V)	2000	2000 ... 1700 ¹⁾	2000 ... 1000 ¹⁾
P_{St} (in kVA)	see diagram P_{St} / I_U m		

Table 2 Rated through-currents (I_U), related rated step voltages (U_i) and step capacities (P_{St}), at highest and lowest rated step voltage of the step capacity curve (see **fig. 3**).

Note: ¹⁾ See also diagram $U_i - I_U$.

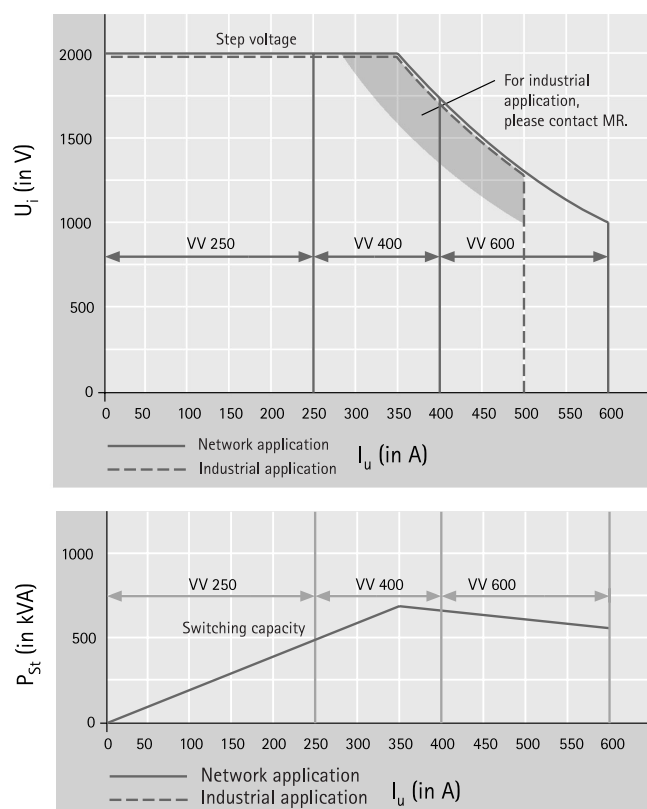


Fig. 3 Step capacities (I_U , U_i)

2.2 Life of the vacuum interrupters

The vacuum interrupters should be replaced as a precautionary measure after 600,000 tap change operations.

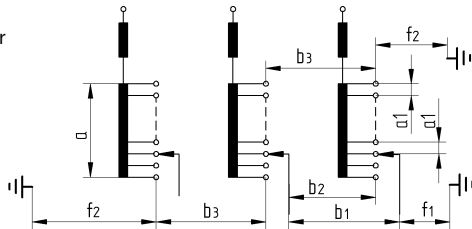
2 Technical Data

2.3 Rated withstand voltages of the internal insulation

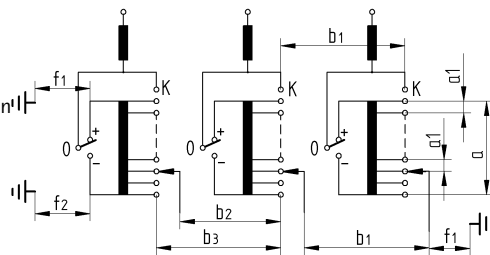
Insulation distances		Voltage shape and period		VV III xxx D		VV III xxx Y		VVI			
a	U _m = 40 kV	kV 1.2/50 μs	200		200		U _m = 76 kV	200			
	U _m = 76 kV	kV 50Hz 1 min	50		50			50			
	U _m = 145 kV	kV 1.2/50 μs	250			U _m = 145 kV	250				
		kV 50Hz 1 min	65				65				
a1	U _m = 40 kV	kV 1.2/50 μs	100		100		U _m = 76 kV	100			
	U _m = 76 kV	kV 50Hz 1 min	20		20			20			
	U _m = 145 kV	kV 1.2/50 μs	130			U _m = 145 kV	130				
		kV 50Hz 1 min	30				30				
b			b ₁	b ₂	b ₃						
	U _m = 40 kV	kV 1.2/50 μs	200	250	300	200					
		kV 50Hz 1 min	70	80	90	85					
	U _m = 76 kV	kV 1.2/50 μs	350	490	520	200					
		kV 50Hz 1 min	140	165	180	85					
	U _m = 145 kV	kV 1.2/50 μs	650	730	800						
		kV 50Hz 1 min	275	285	300						
	c1	U _m = 40 kV	kV 1.2/50 μs	400		400					U _m = 76 kV
U _m = 76 kV		kV 50Hz 1 min	100		100		100				
U _m = 145 kV		kV 1.2/50 μs	500			U _m = 145 kV	500				
		kV 50Hz 1 min	130				130				
c2			c2 ₁	c2 ₂							
	U _m = 40 kV	kV 1.2/50 μs	250	300		350					
		kV 50Hz 1 min	80	90		140					
	U _m = 76 kV	kV 1.2/50 μs	490	520		350					
		kV 50Hz 1 min	165	180		140					
	U _m = 145 kV	kV 1.2/50 μs	730	800							
		kV 50Hz 1 min	285	300							
	f			f ₁	f ₂						
U _m = 40 kV		kV 1.2/50 μs	200	250		200					
		kV 50Hz 1 min	70	80		70					
U _m = 76 kV		kV 1.2/50 μs	350	490		350		U _m = 76 kV	350	490	
		kV 50Hz 1 min	140	165		140			140	165	
U _m = 145 kV		kV 1.2/50 μs	650	730			U _m = 145 kV	650	730		
		kV 50Hz 1 min	275	285				275	285		

Table 3 Rated withstand voltages (in kV) of the insulation distances (rated lightning impulse withstand voltage, AC withstand voltage) on the on-load tap-changer.

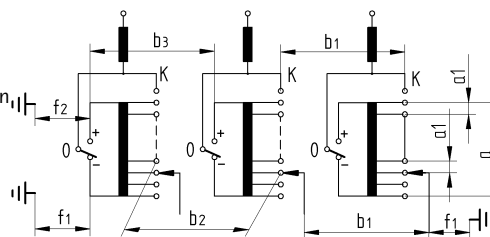
Without
change-over
selector



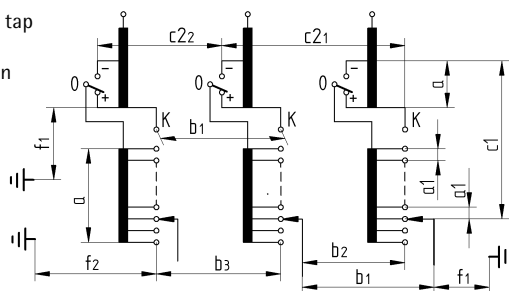
With
reversing
switch
(+) - position



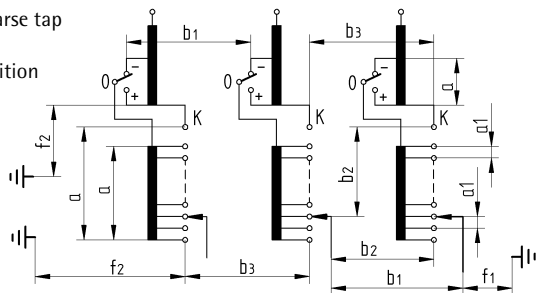
With
reversing
switch
(-) - position



With coarse tap
selector
(+) - position



With coarse tap
selector
(-) - position



8997580E

Fig. 4
Specific dielectric stress of the tap winding

a1=Between tap selector contacts of the winding of one step position (connected or not).

a =Between beginning and end of a tapped winding and, with coarse tap winding, between beginning and end of a coarse tap winding. Note for coarse tapping arrangement in (-) -position of the change-over selector:

When stressed with impulse voltage, the permissible withstand voltage "a" must be adhered to between the end of a coarse tap winding connected with the K tap selector contact and the tap selector contact at the end of the tapped winding of the same phase.

b =Between the tap selector contacts of different phases and between change-over selector contacts of different phases, which are connected with the beginning/end of a tapped winding or with a tap selector contact.

The permissible voltages between the contacts for tap-changers for delta connection vary with the positions of the change-over selector and the tap selector. This is the reason the different permissible withstand voltages (b_1 , b_2 or b_3) must be adhered to for an insulation distance.

b_1 =Between selected contacts of different phases.

b_2 =From the selected contact of one phase to the unselected contacts of other phases.

b_3 =Between unselected contacts of different phases.

f = Between tap selector contacts and (+) -change-over selector contacts to ground.

Different values apply to designs with delta connection:

f_1 = Between terminal and ground and in the (+) - position of the change-over selector, between (+) -change-over selector contacts and ground.

f_2 = Between unselected tap selector contacts and in the (-) -position of the change-over selector, between (+) - contacts and ground.

Additional for coarse tapping arrangement in (+) - position of the change-over selector:

c1 = From one (-) - change-over selector contact to terminal of the same phase.

c2 = Between (-) -change-over selector contacts of different phases and a (-) - change-over selector contact to a (+) - change-over selector contact of another phase.

Different permissible values apply to tap-changers with delta connection:

c_{2_1} = Between a (-) -change-over selector contact and a (+) - change-over selector contact of another phase.

c_{2_2} = Between (-) change-over selector contacts of different phases.

3 Appendix

3.1 Dimension drawings

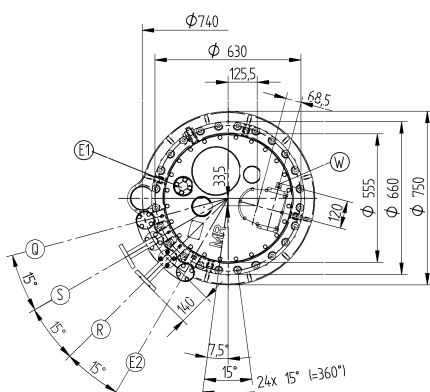
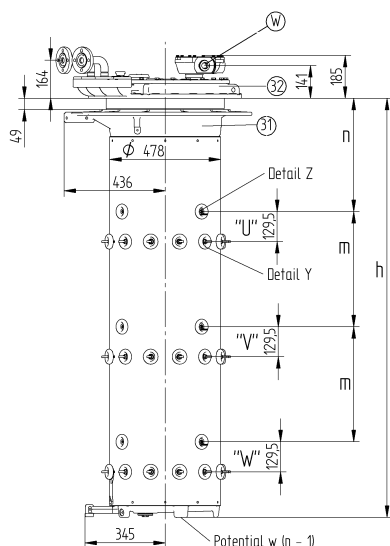
On-load tap-changer VACUTAP® VV III 250/400/600 -Y/D - 40 ...76 -10/12		
Without change-over selector	899120	Page 17
With change-over selector	899119	Page 18
Laststufenschalter VACUTAP® VV III 250/400/600 D-145 -10/12		
Without change-over selector	899163	Page 19
With change-over selector	899162	Page 20
Laststufenschalter VACUTAP® VV I 401-76 -10/12		
With change-over selector	899121	Page 21
Laststufenschalter VACUTAP® VV I 401-145 -10/12		
With change-over selector	727986	Page 22

3.2 Installation drawings

On-load tap-changer VACUTAP® VV	898864	Page 23
On-load tap-changer VACUTAP® VV, tie-in resistors, installed on the side	899407	Page 24
On-load tap-changer VACUTAP® VV, mounted for bell-type tank installation	899110	Page 25
On-load tap-changer VACUTAP® VV, location of the tap-changer for bell-type tank installation	899409	Page 26
On-load tap-changer VACUTAP® VV, location of the terminal contacts	899051	Page 27
On-load tap-changer VACUTAP® VV, mounted for cover installation	898866	Page 28

3.3 Additional drawings

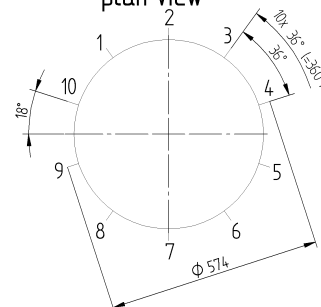
On-load tap-changer VACUTAP® VV, on-load tap-changer head	898863	Page 29
On-load tap-changer VACUTAP® VV, special design with flange for pressure relief valve	899764	Page 30
On-load tap-changer VACUTAP® VV, special design with flange for pressure relief valve	899210	Page 31
On-load tap-changer VACUTAP® VV, special design with flange for pressure relief valve	899211	Page 32
On-load tap-changer VACUTAP® VV, horizontal drive shaft	899169	Page 33



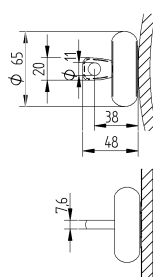
- (E1) - Bleeding facility for Tap-Changer head
- (E2) - Bleeding facility for space under the head outside the oil compartment
- (Q) - Pipe connection (optional)
- (R) - Connection for protective relay
- (S) - Pipe connection
- (W) - Drive shaft
- (31) - Tap-Changer head - bottom part
- (32) - Tap-Changer head top part
(can be turned towards the bottom part of the head in steps of 15°)
(Tap-Changer head cover cannot be turned)

U _m [kV]		40	76
Dimensions [mm]	m	432	497
	n	437	489
	h	1628	1810

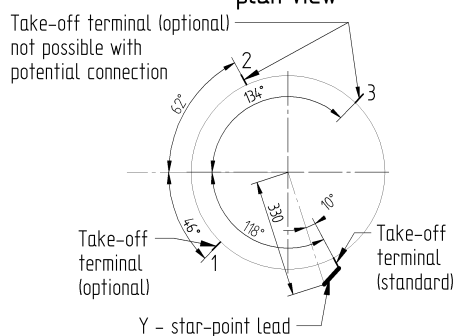
10 pitch contact arrangement
plan view



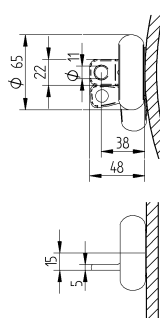
Y
Fine-tap selector
terminal Ta



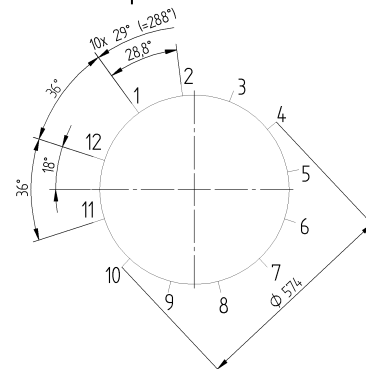
Location of 10 pick take-off terminals
plan view



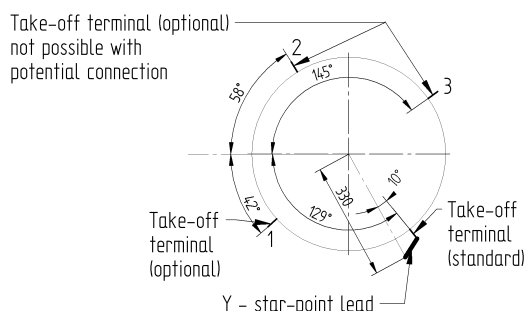
Z
Take-off terminal



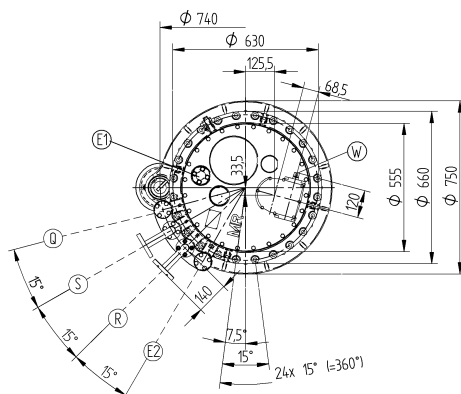
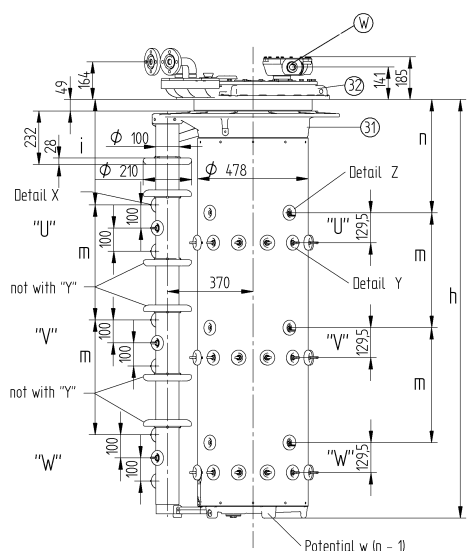
12 pitch contact arrangement
plan view



Location of 12 pitch take-off terminals
Plan view



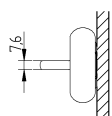
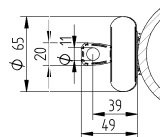
The connection diagram is binding for the designation of the terminals and phases.



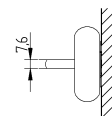
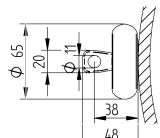
- ①E1 – Bleeding facility for Tap-Changer head
- ①E2 – Bleeding facility for space under the head outside the oil compartment
- ①Q – Pipe connection (optional)
- ①R – Connection for protective relay
- ①S – Pipe connection
- ①W – Drive shaft
- ③1 – Tap-Changer head - bottom part
- ③2 – Tap-Changer head - top part
(can be turned towards the bottom part of the head in steps of 15°)
the Tap-Changer head cover cannot be turned

U _m [kV]		40	76
Dimensions [mm]	i	404	456
	m	432	497
	n	437	489
	h	1628	1810

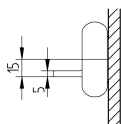
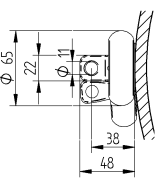
X
Change-over
selector terminal



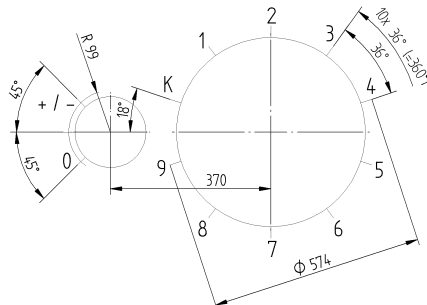
Y
Fine tap
selector terminal



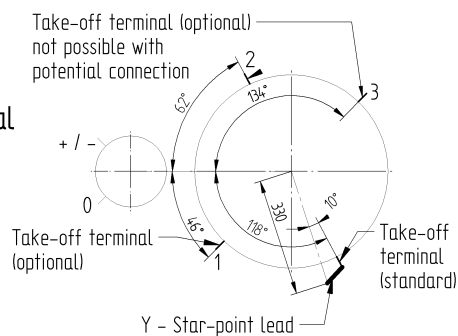
Z
Take-off terminal



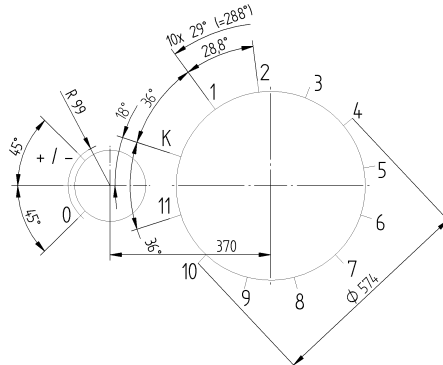
10 pitch contact arrangement
Plan view



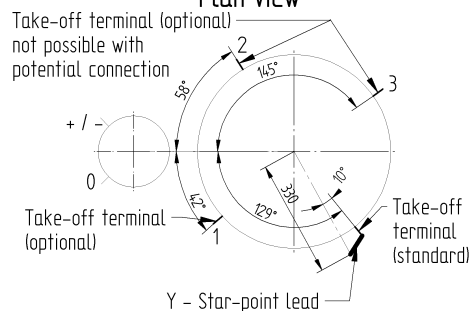
Location of 10 pitch take-off terminals
Plan view



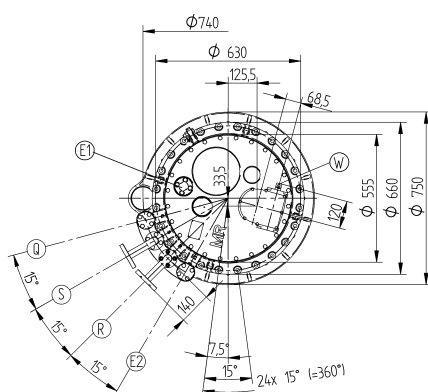
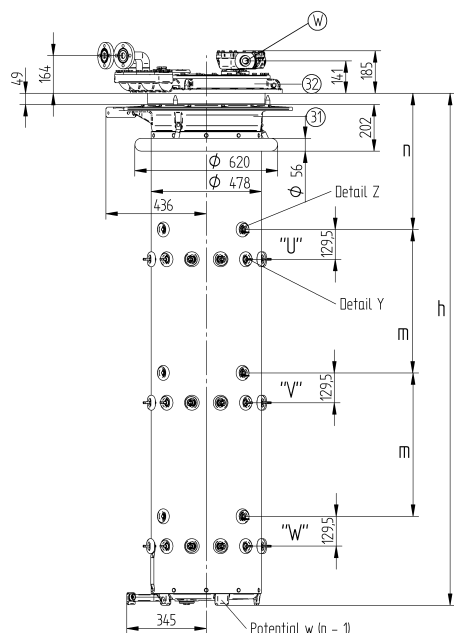
12 pitch contact arrangement
Plan view



Location of 12 pitch take-off terminals
Plan view

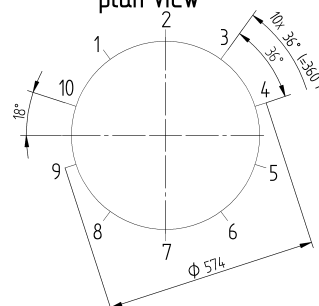
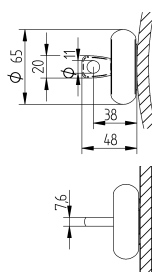
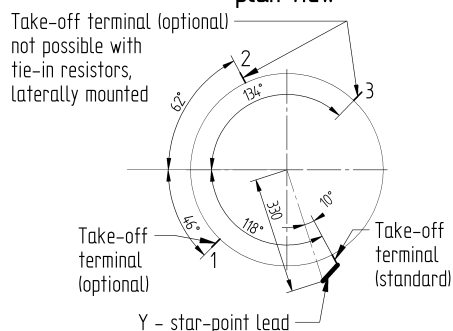
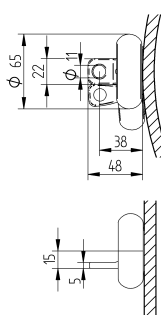
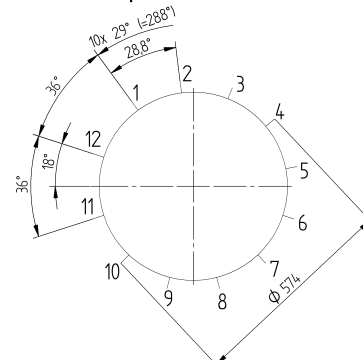
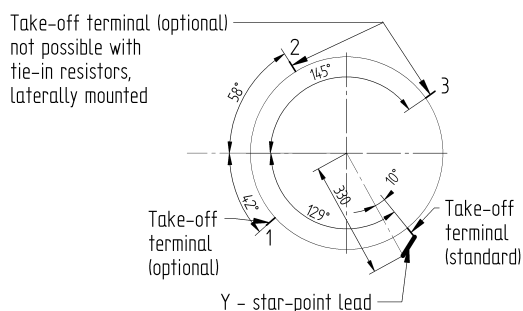


The connection diagram is binding for the designation of the terminals and phases

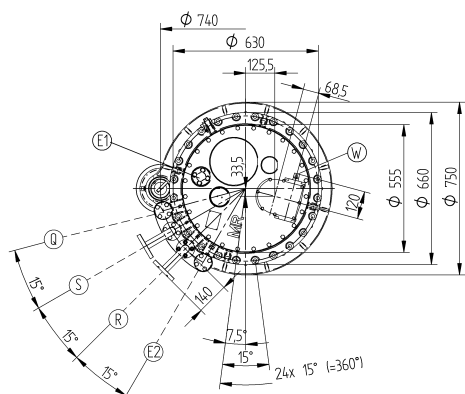
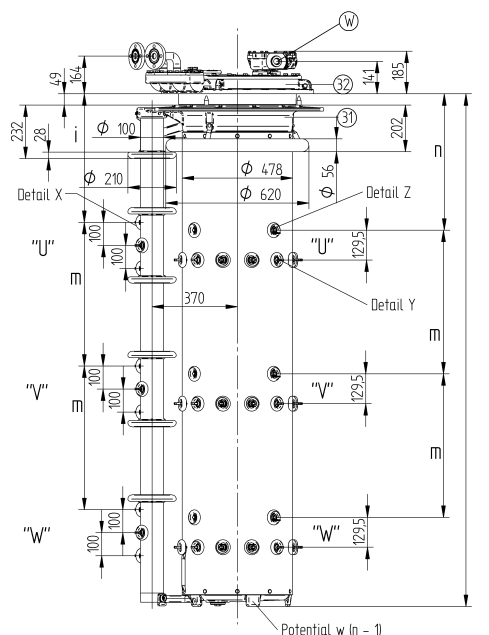


- (E1) - Bleeding facility for Tap-Changer head
- (E2) - Bleeding facility for space under the head outside the oil compartment
- (Q) - Pipe connection (optional)
- (R) - Connection for protective relay
- (S) - Pipe connection
- (W) - Drive shaft
- (31) - Tap-Changer head - bottom part
- (32) - Tap-Changer head top part (can be turned towards the bottom part of the head in steps of 15°) (Tap-Changer head cover cannot be turned)

U _m [kV]	145
Dimensions [mm]	m 623
	n 591
	h 2224

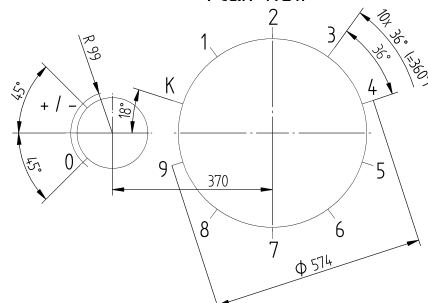
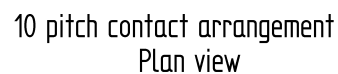
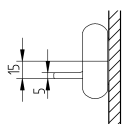
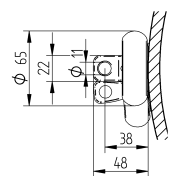
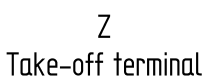
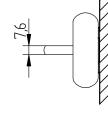
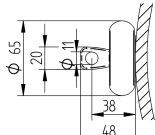
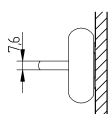
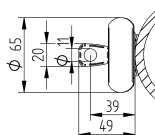
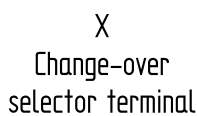
10 pitch contact arrangement
plan viewY
Fine-tap selector
terminalLocation of 10 pitch take-off terminals
plan viewZ
Take-off terminal12 pitch contact arrangement
plan viewLocation of 12 pitch take-off terminals
Plan view

The connection diagram is binding for the designation of the terminals and phases.

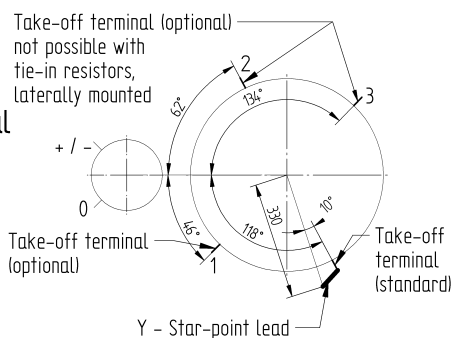


- (E1) – Bleeding facility for Tap-Changer head
- (E2) – Bleeding facility for space under the head outside the oil compartment
- (Q) – Pipe connection (optional)
- (R) – Connection for protective relay
- (S) – Pipe connection
- (W) – Drive shaft
- (31) – Tap-Changer head - bottom part
- (32) – Tap-Changer head - top part
(can be turned towards the bottom part of the head in steps of 15°)
the Tap-Changer head cover cannot be turned

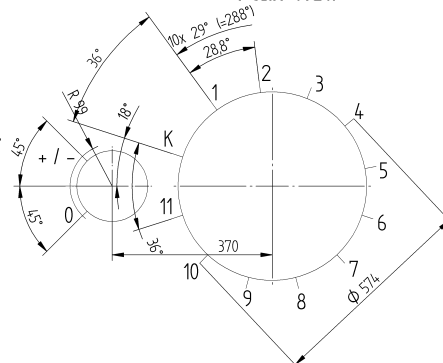
U _m [kV]		145
Dimension [mm]	i	558
	m	623
	n	591
	h	2224



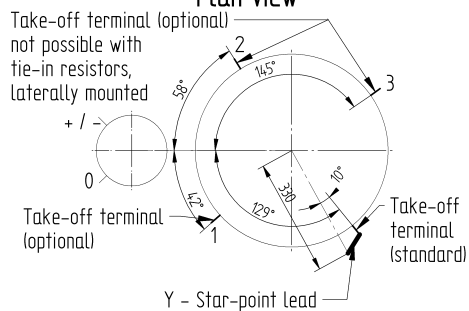
Location of 10 pitch take-off terminals
Plan view



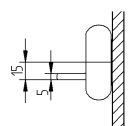
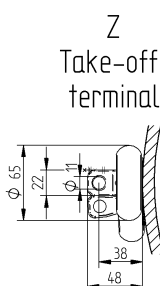
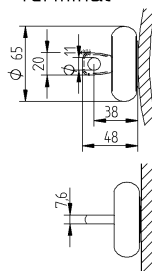
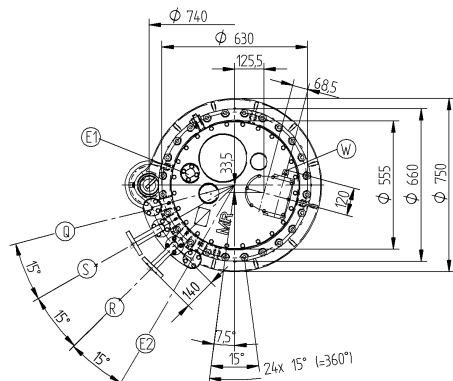
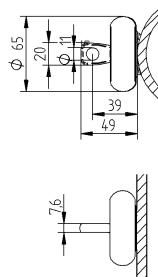
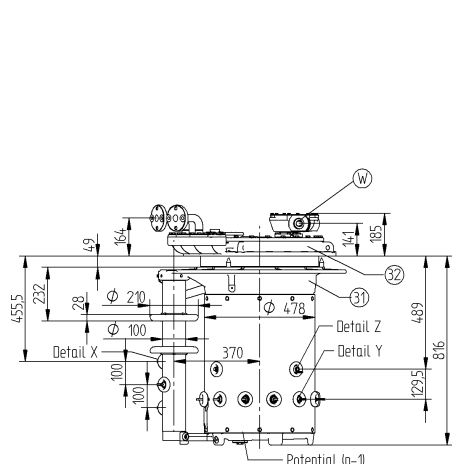
12 pitch contact arrangement
Plan view



Location of 12 pitch take-off terminals
Plan view

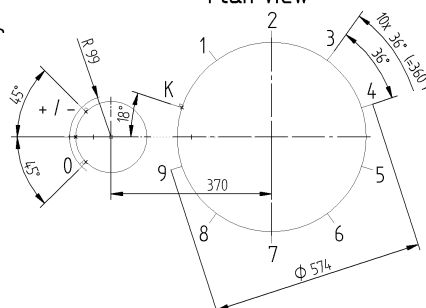


The connection diagram is binding for the designation of the terminals and phases

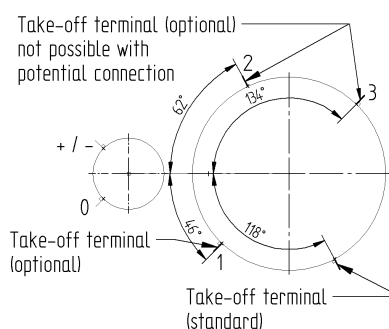


- (E1) - Bleeding facility for Tap-Changer head
- (E2) - Bleeding facility for space under the head outside the oil compartment
- (Q) - Pipe connection (optional)
- (R) - Connection for protective relay
- (S) - Pipe connection
- (W) - Drive shaft
- (31) - Tap-Changer head - bottom part
- (32) - Tap-Changer head top part
(can be turned towards the bottom part of the head in steps of 15°)
(Tap-Changer head cover cannot be turned)

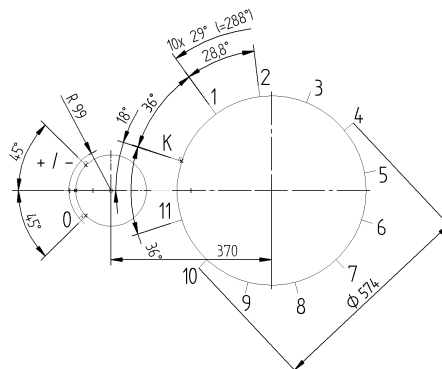
10 pitch contact arrangement
Plan view



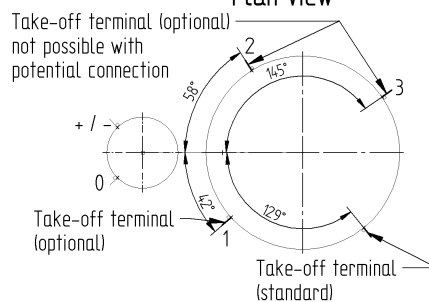
Location of 10 pich take-off terminals
Plan view



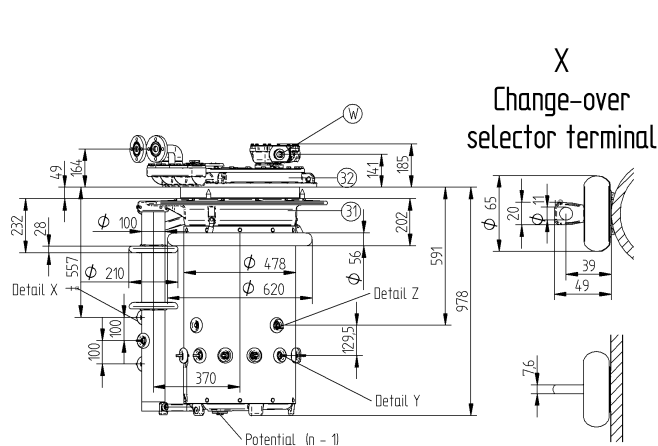
12 pitch kontakt arrangement
Plan view



Location of 12 pitch take-off terminals
Plan view

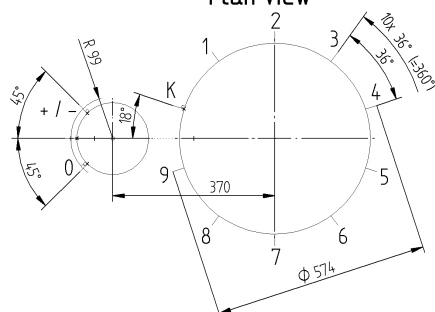


The connection diagram is binding for the designation of the terminals and phases.

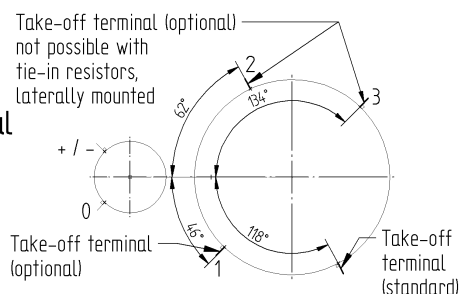


X
Change-over
selector terminal

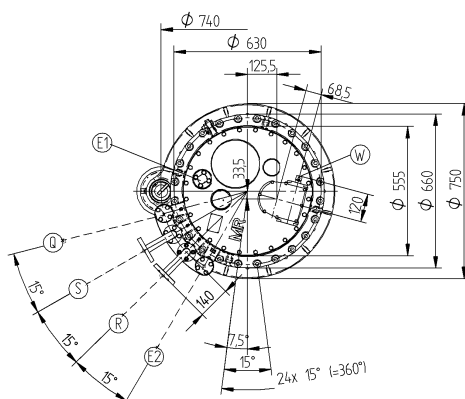
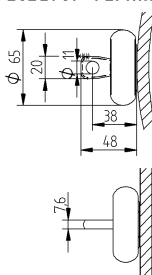
10 pitch contact arrangement
Plan view



Location of 10 pitch take-off terminals
Plan view

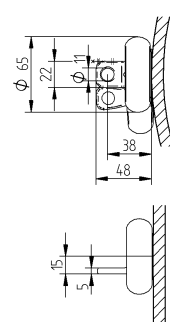


Y
Fine tap
selector terminal

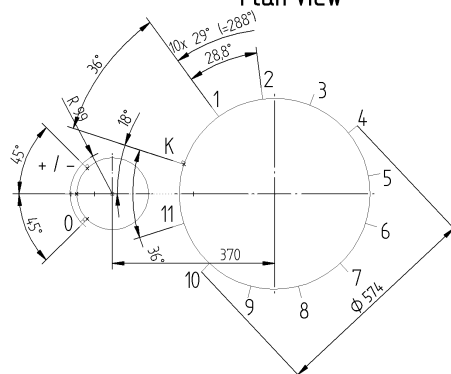


- (E1) + Bleeding facility for Tap-Changer head
- (E2) + Bleeding facility for space under the head outside the oil compartment
- (Q) + Pipe connection (optional)
- (R) + Connection for protective relay
- (S) + Pipe connection
- (W) + Drive shaft
- (31) + Tap-Changer head - bottom part
- (32) + Tap-Changer head - top part
(can be turned towards the bottom part of the head in steps of 15°)
the Tap-Changer head cover cannot be turned

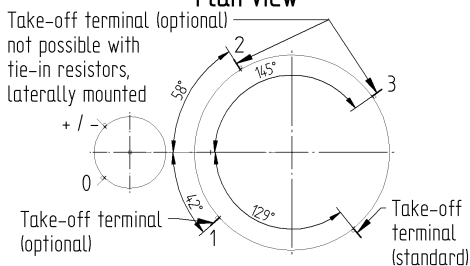
Z
Take-off terminal



12 pitch contact arrangement
Plan view



Location of 12 pitch take-off terminals
Plan view



The connection diagram is binding for the designation of the terminals and phases

The connection diagram is binding
for the designation of the
terminals and phases

- 11 Change-over selector pipe
- 12 Change-over selector terminal "+"
- 13 Change-over selector terminal "0"
- 14 Change-over selector terminal "-"

- 21 Take-off terminal
- 22 Fine tap selector terminal

- 23 Oil compartment bottom with kerosene drain plug 23a
- 24 Tap changer oil compartment

- 31 Tap changer head - bottom part
- 32 Tap changer head - top part

- 33 Pipe connection E2 (optional)

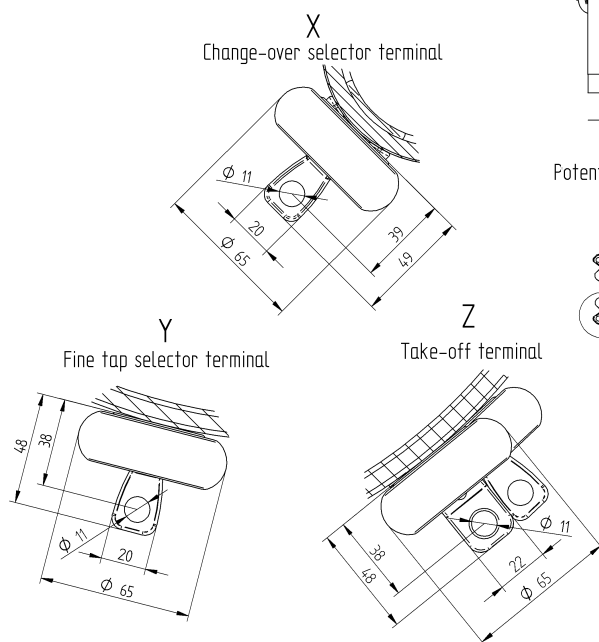
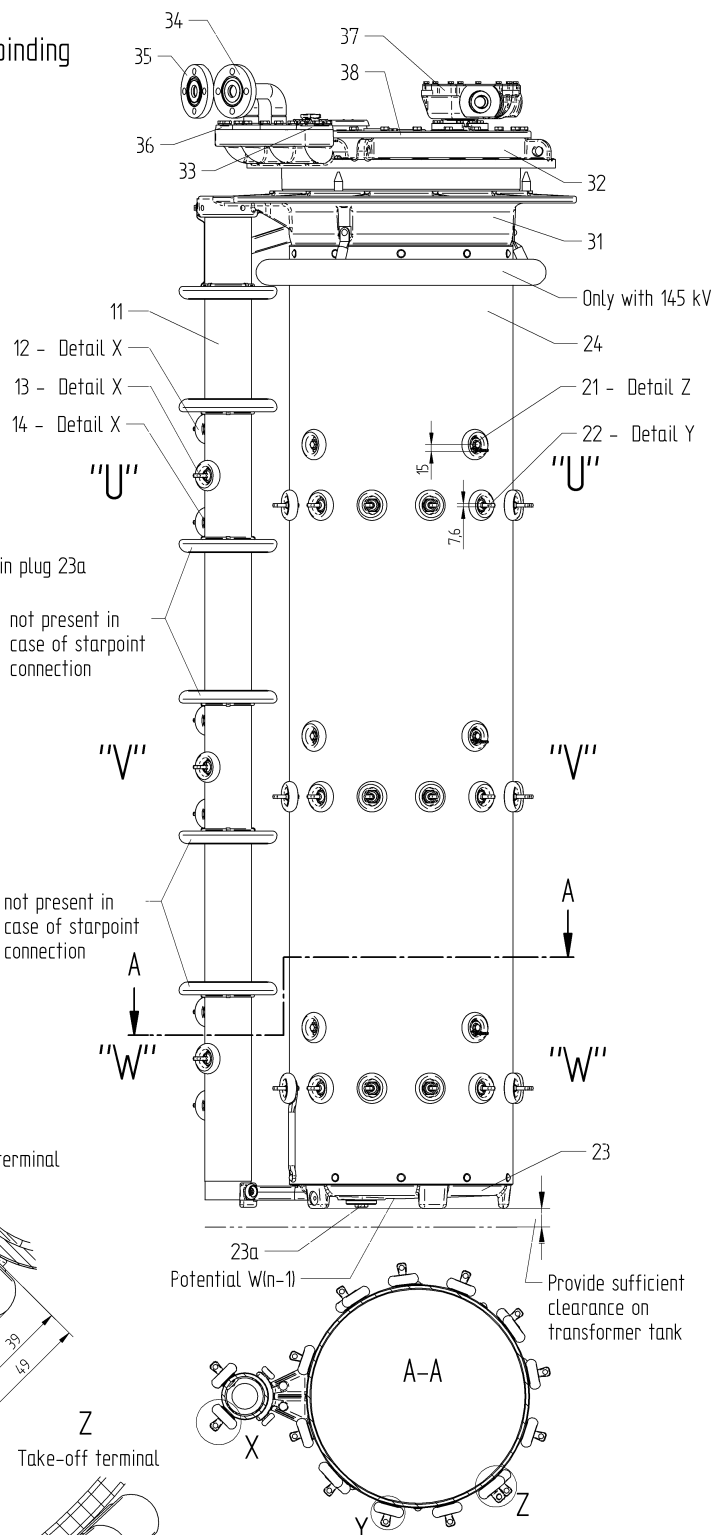
- 34 Pipe connection R for protective relay

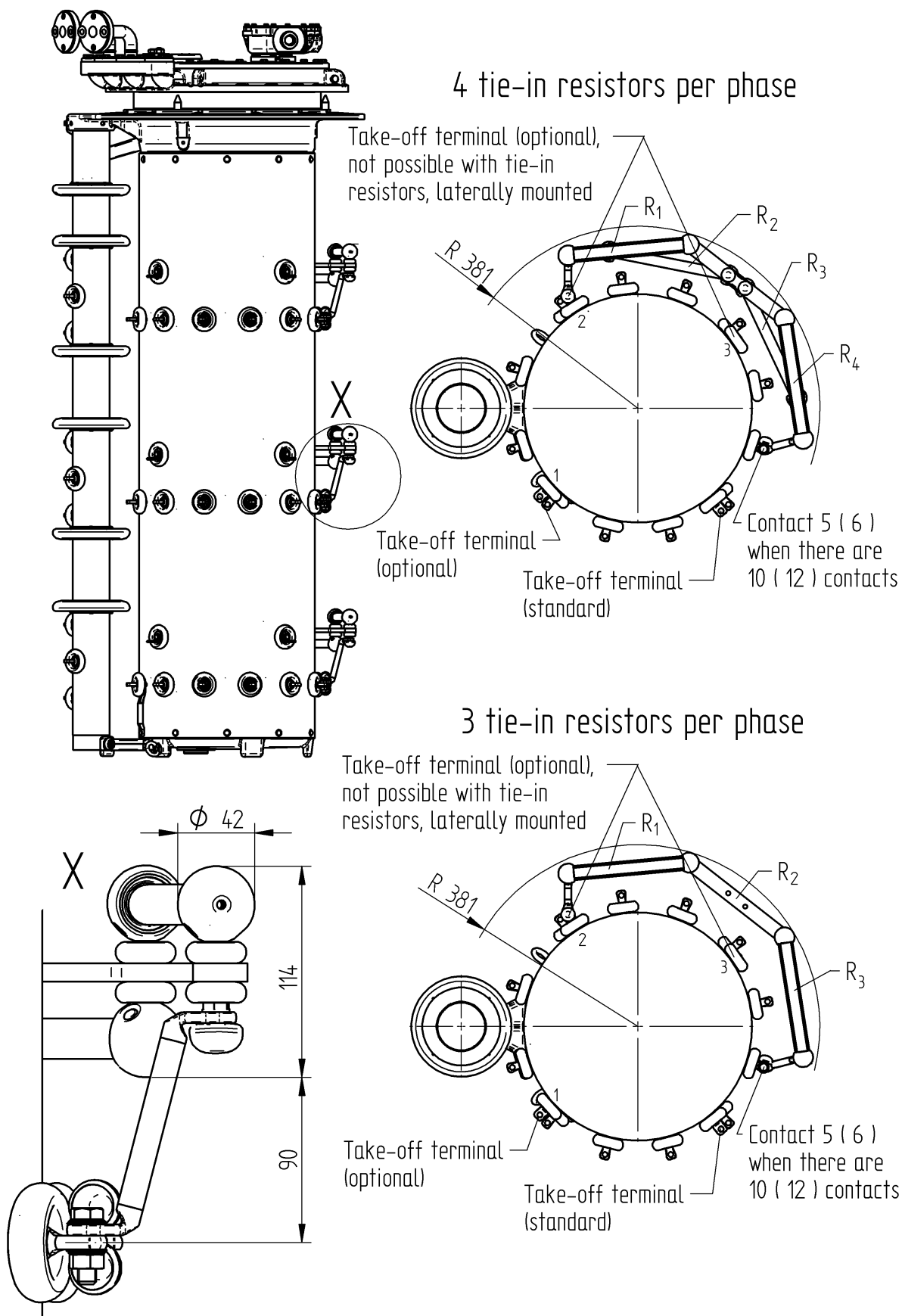
- 35 Pipe connection S

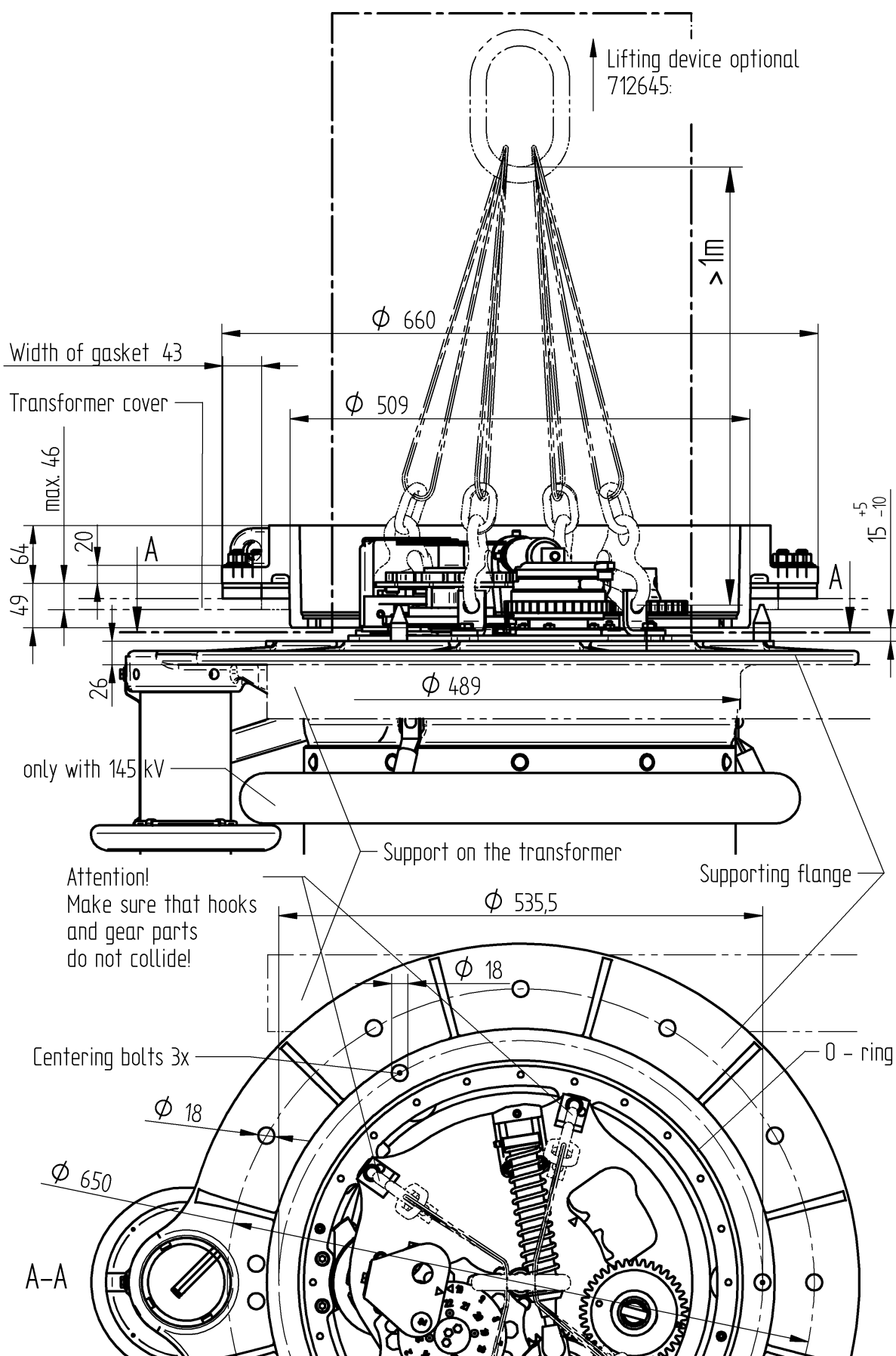
- 36 Pipe connection Q (optional)

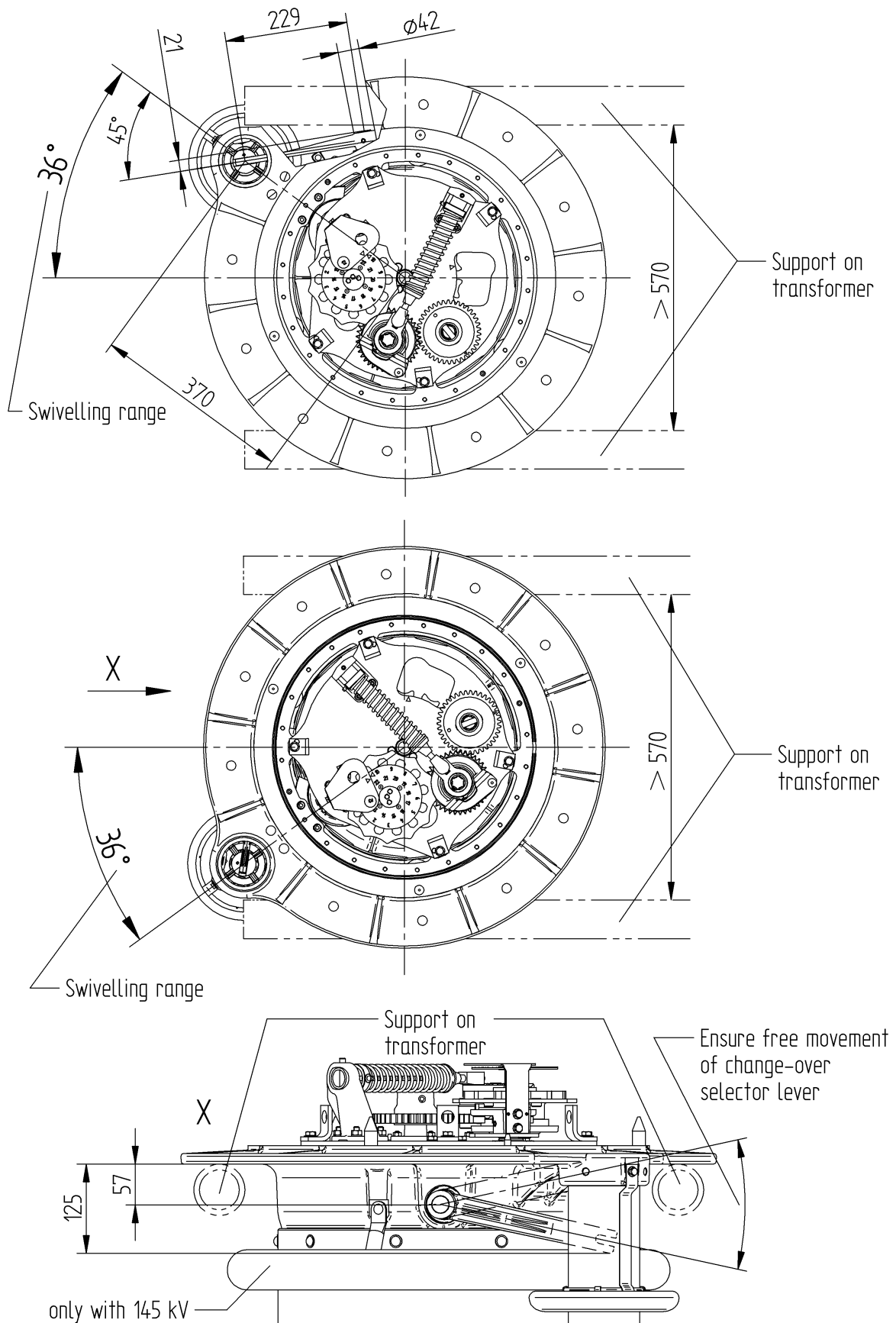
- 37 Upper gear unit with drive shaft

- 38 Tap changer head cover

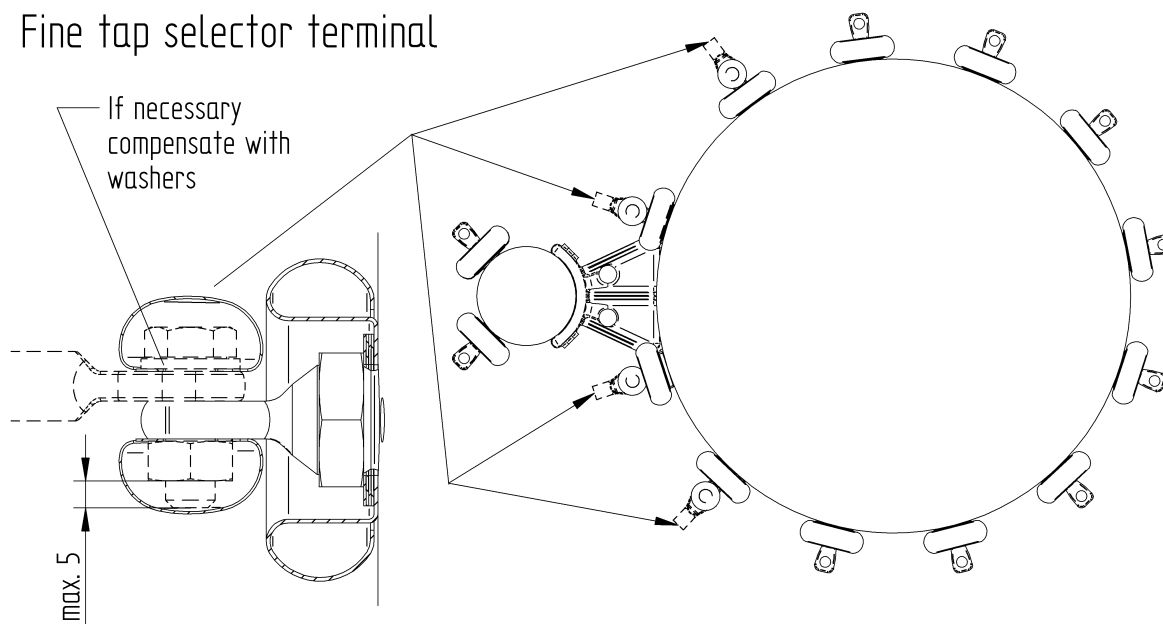








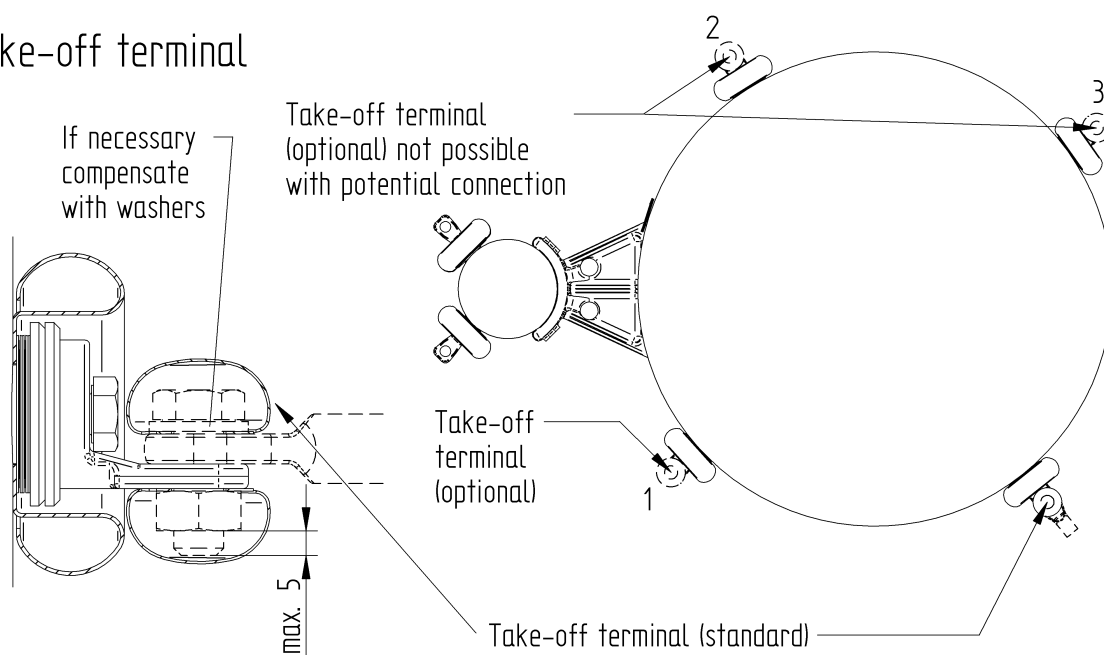
Fine tap selector terminal



These terminals are to be fitted with screening caps (included in MR's delivery) by the transformer manufacturer.

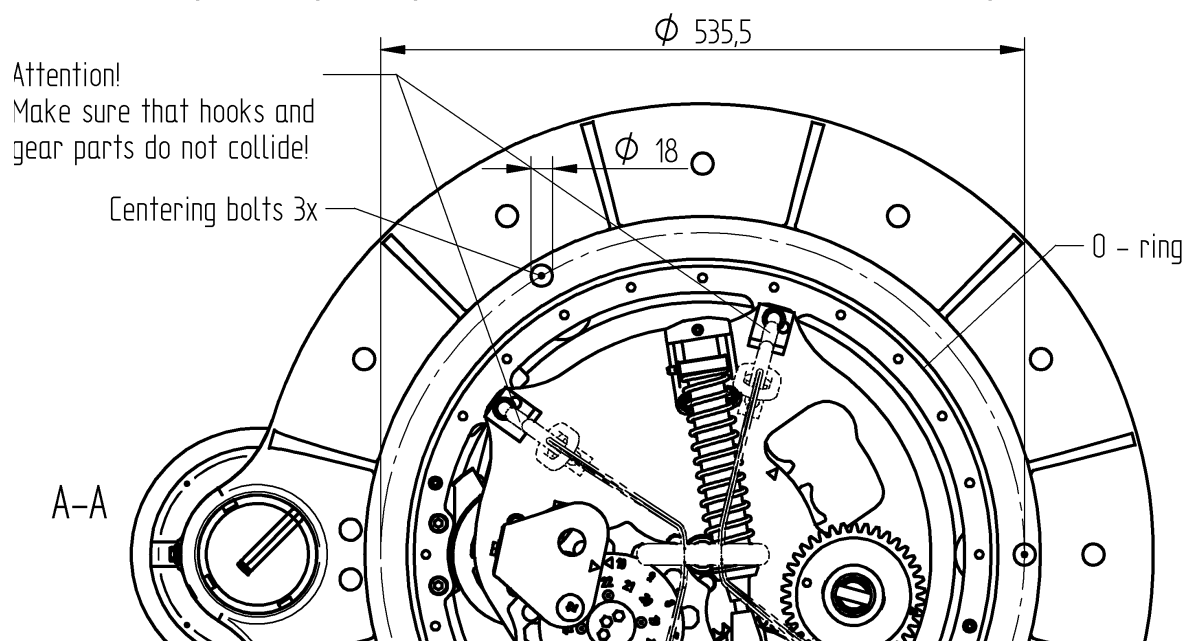
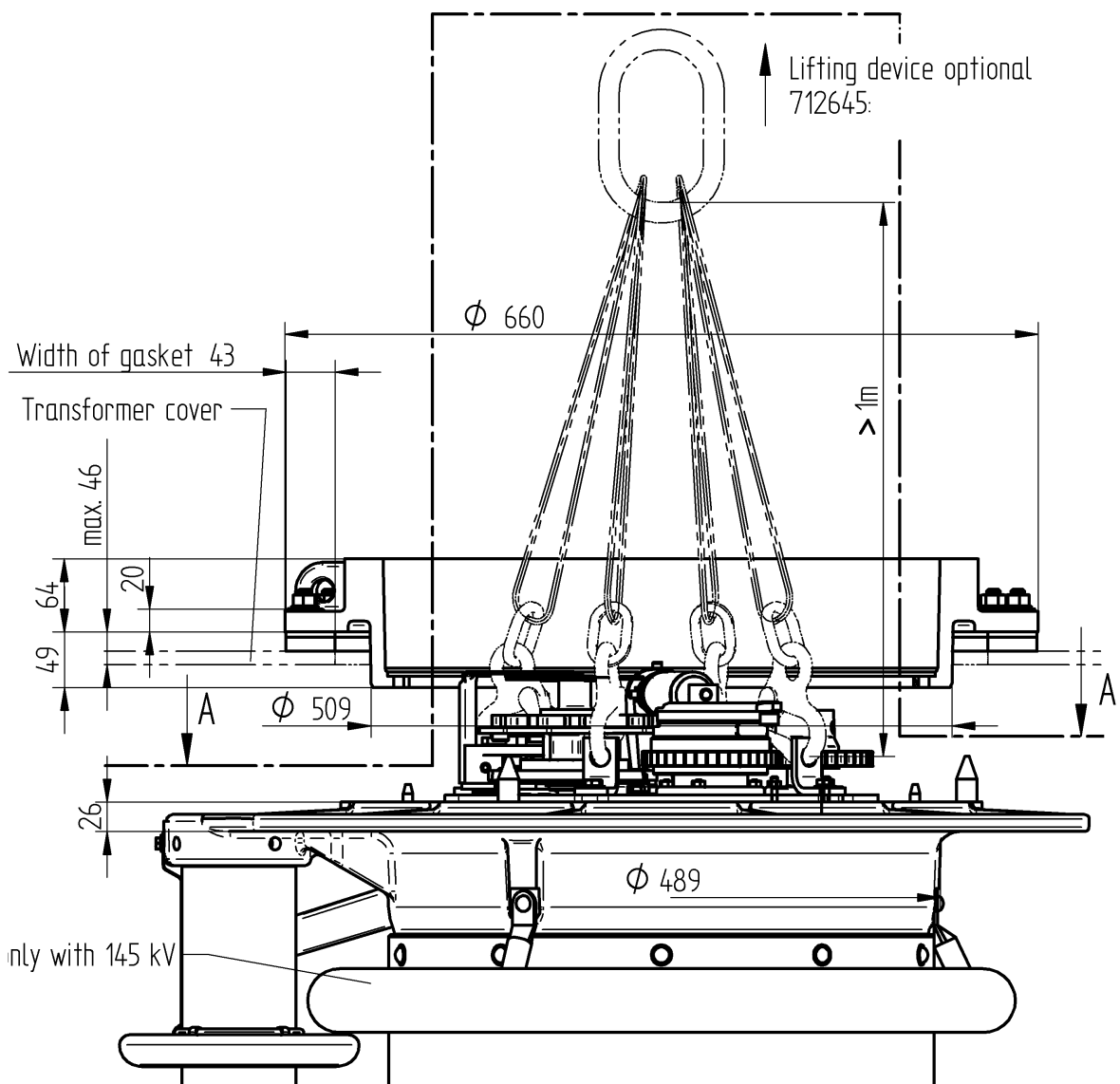
The connection diagram is binding for the designation of the terminals and phases.

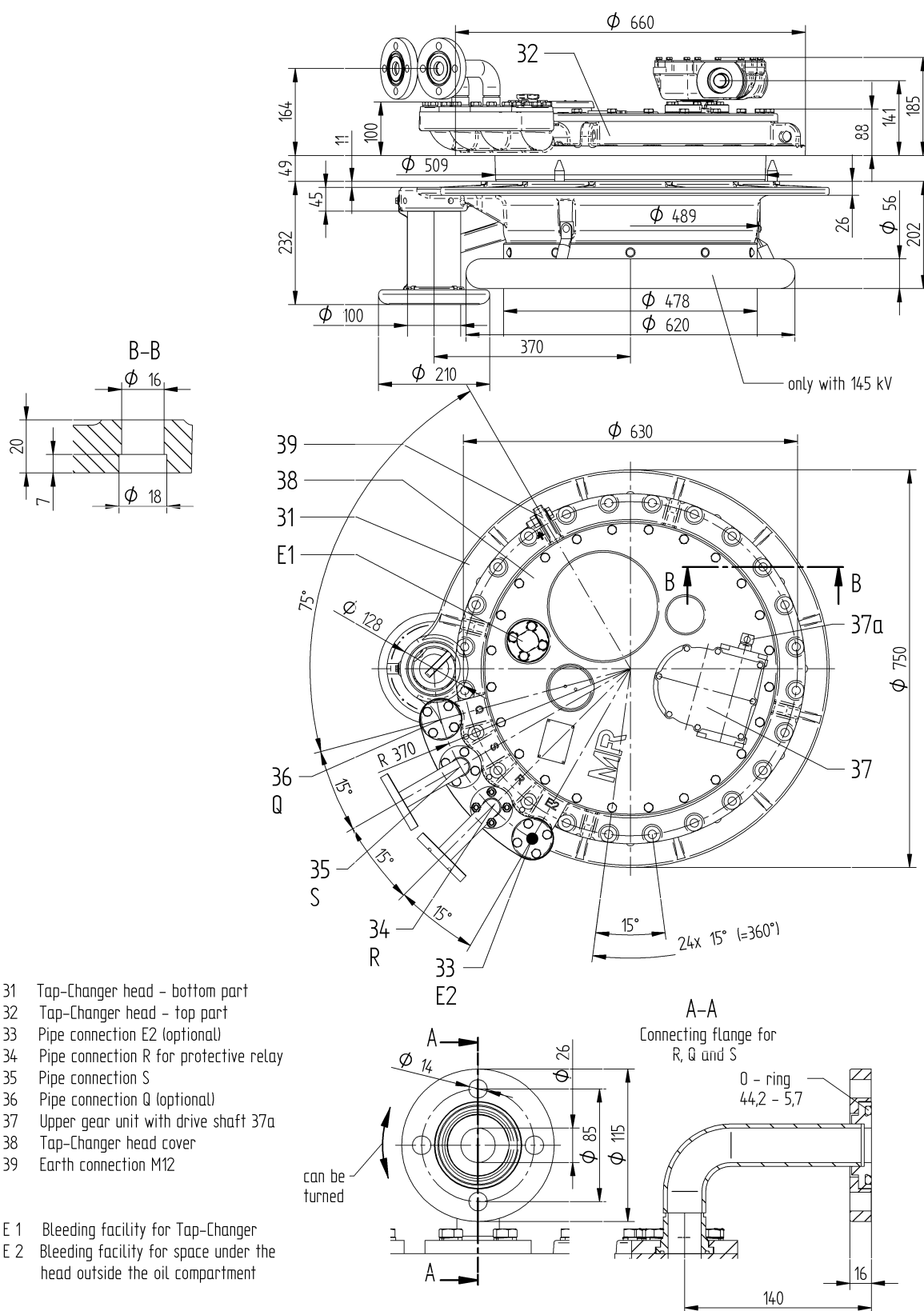
Take-off terminal

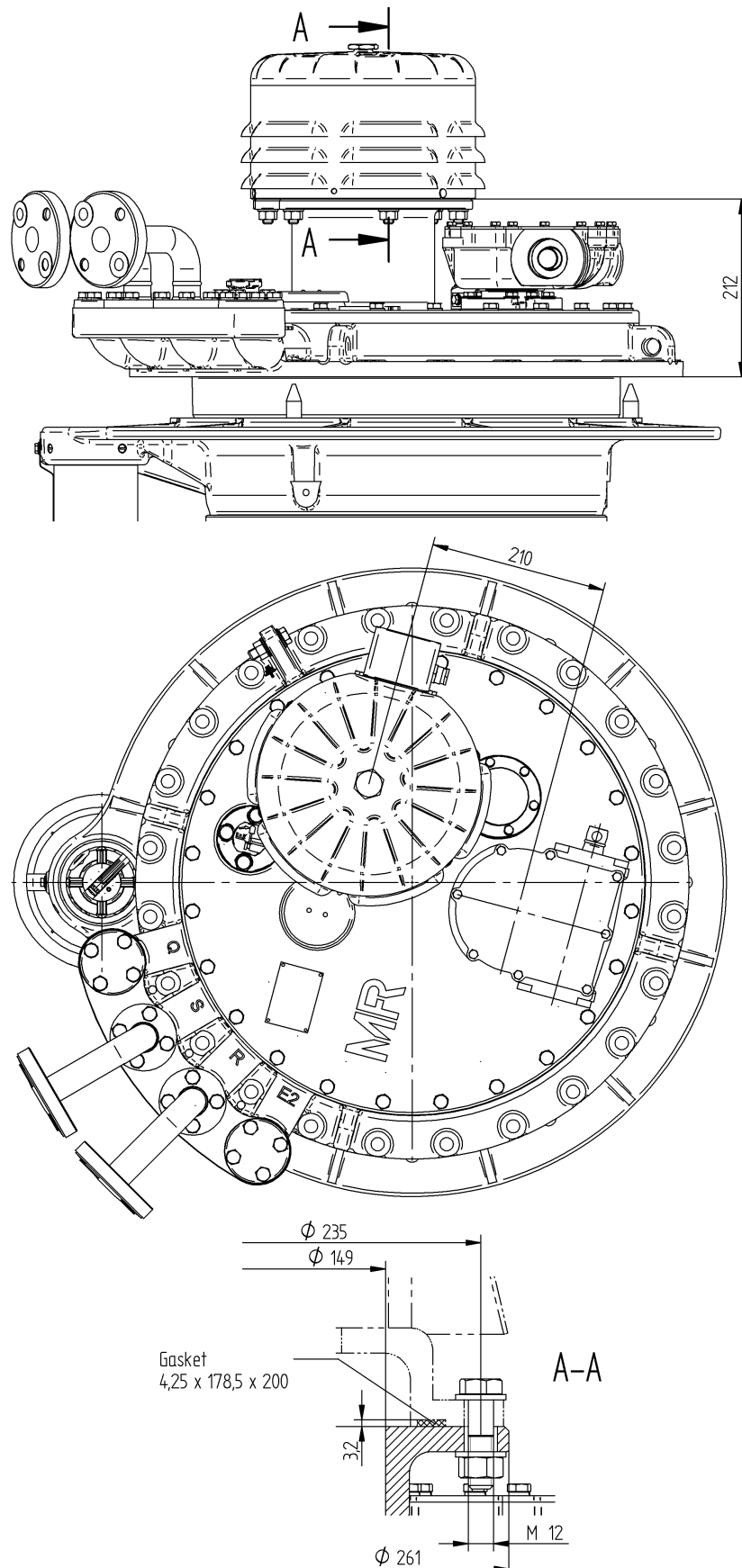


These take-off terminals (with or without take-off leads) are to be fitted with screening caps (included in MR's delivery) by the transformer manufacturer.

Drawing shows on-load tap-changer V V in 12 pitch design

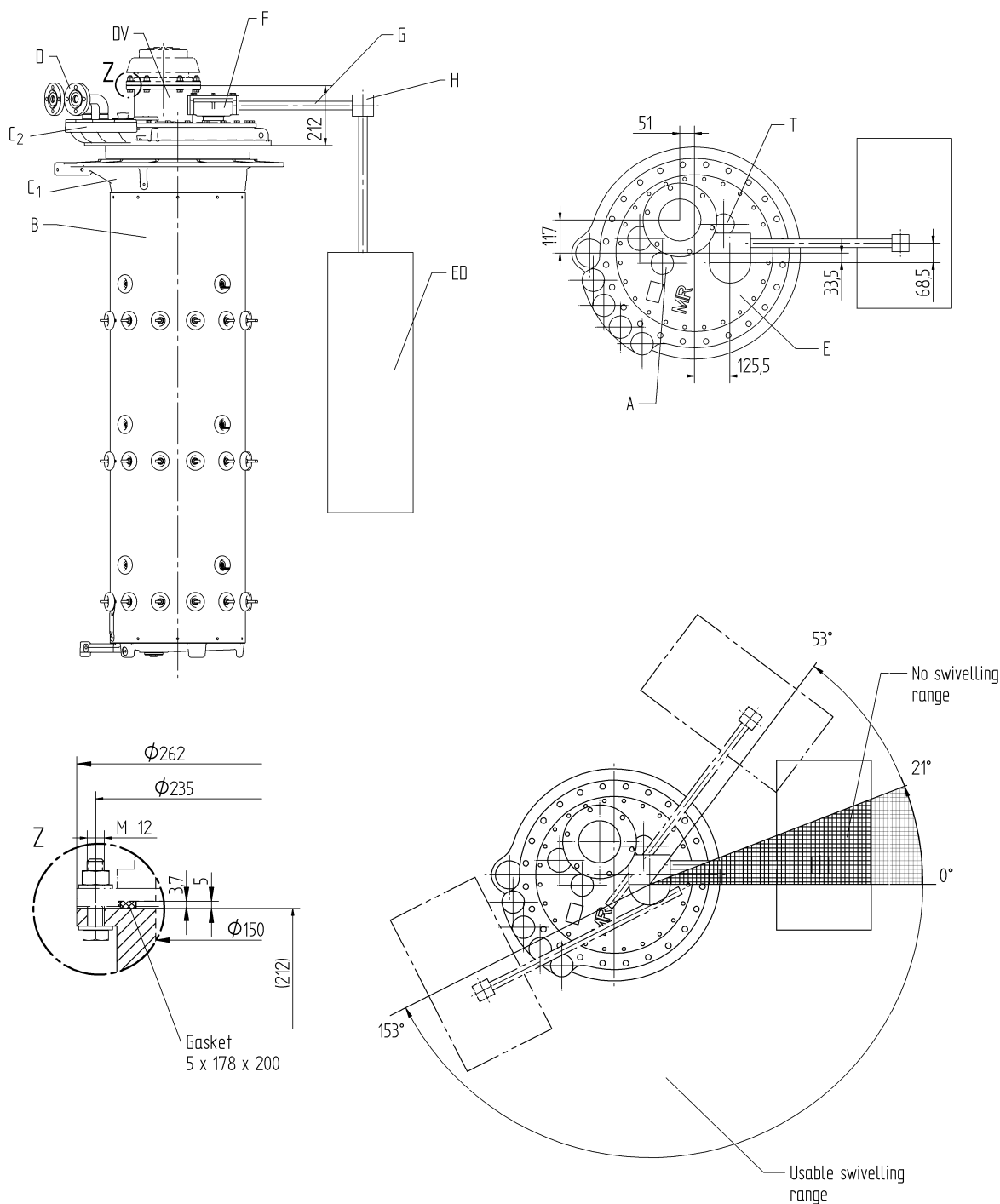






On-load tap-changer VACUTAP® VV

Special design with flange for pressure relief valve
Swivelling range of the gear unit, drive shaft left



- A — Position indicator
- B — Diverter switch oil compartment
- C₁ — Diverter switch head - bottom part
- C₂ — Diverter switch head - top part
- D — Pipe connections R and S
- DV — Flange for pressure relief valve
- E — Tap changer head - cover
- F — Upper gear unit
- G — Horizontal drive shaft
- H — Bevel gear
- ED — Motor drive
- T — Temperature sensor

Swivelling ranges

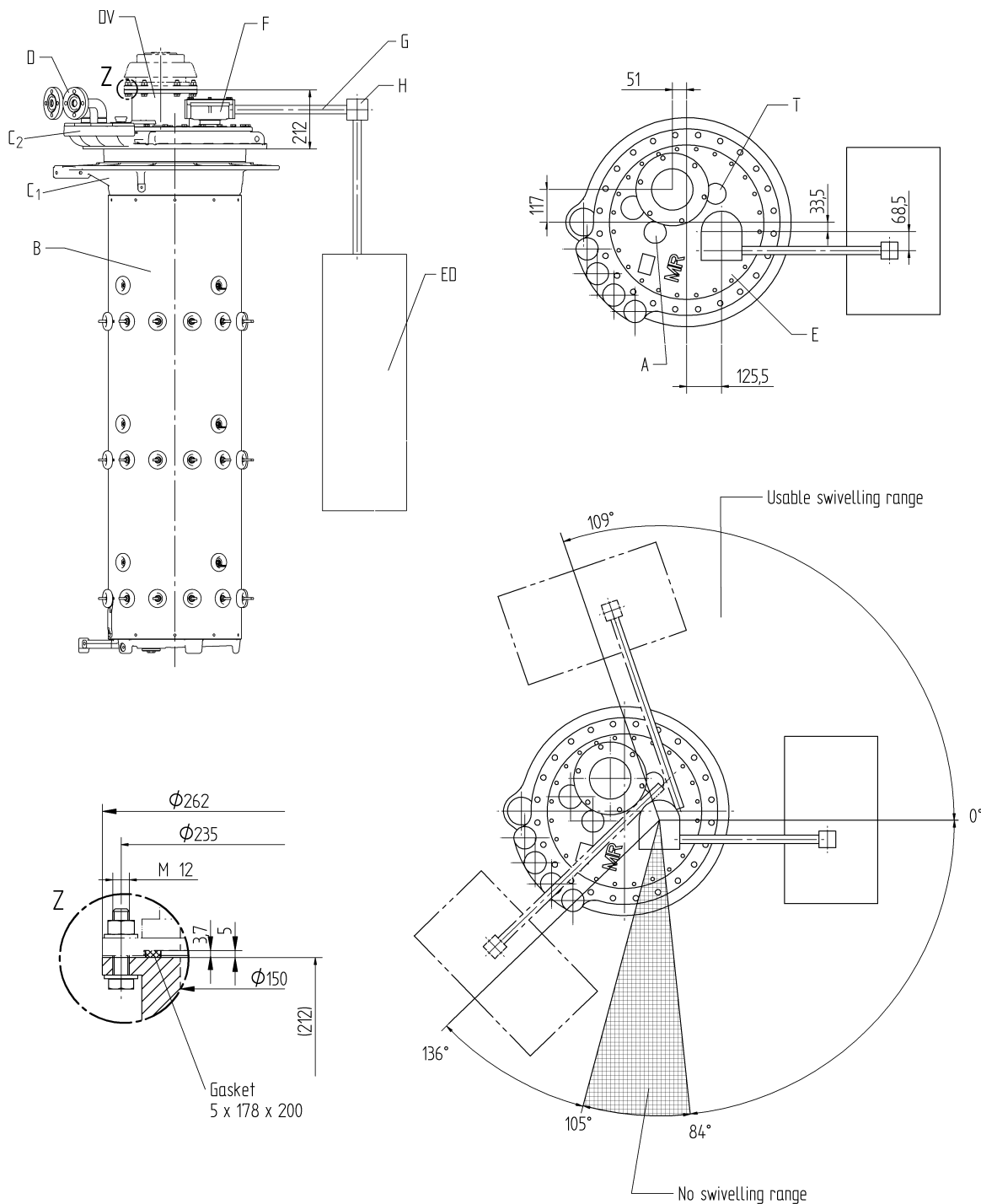
By turning about its own axis and by the steplessly turnable upper gear unit, the tap changer can be well adapted to the transformer.

The usable swivelling ranges of the upper gear unit are shown above.

As the top part and the bottom part of the head can be turned to each other (in steps of 15°), the swivelling ranges change accordingly.

On-load tap-changer VACUTAP® VV

Special design with flange for pressure relief valve
Swivelling range of the gear unit, drive shaft right



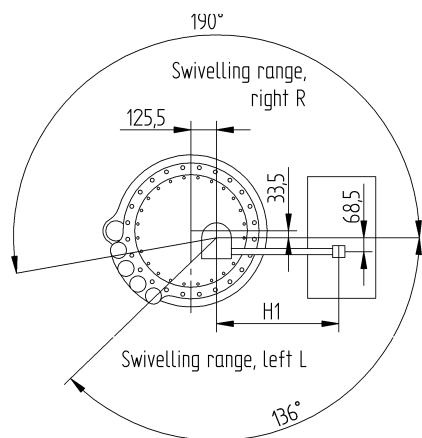
- A — Position indicator
- B — Diverter switch oil compartment
- C₁ — Diverter switch head – bottom part
- C₂ — Diverter switch head – top part
- D — Pipe connections R and S
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- E — Tap changer head – cover
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- G — Horizontal drive shaft
- H — Bevel gear
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Swivelling ranges

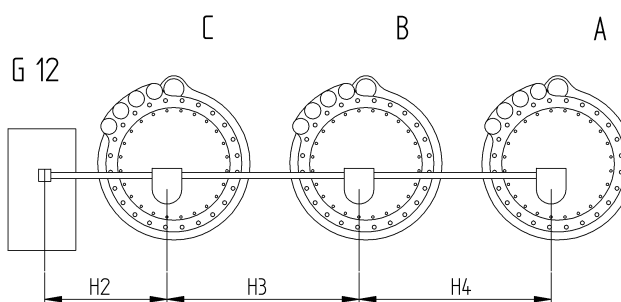
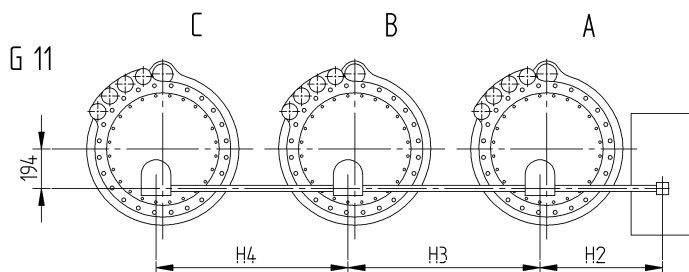
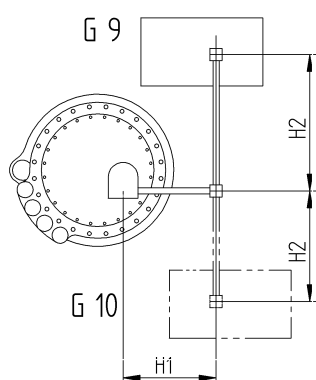
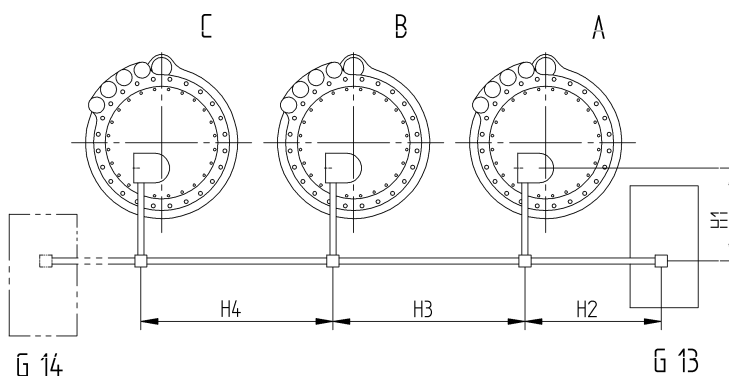
By turning about its own axis and by the steplessly turnable upper gear unit, the tap changer can be well adapted to the transformer.

The usable swivelling ranges of the upper gear unit are shown above.

As the top part and the bottom part of the head can be turned to each other (in steps of 15°), the swivelling ranges change accordingly.

Arrangement G 4
(Standard design)

As the top part of the head can be turned towards the bottom part of the head in 15° steps, the swivelling ranges change accordingly.

Arrangement G 11, G 12
(Standard design)Arrangement G 9, G 10
(Special design)Arrangement G 13, G 14
(Special design)

Minimum dimensions				Intermediate bearing required for		
Arrangement	H1	H2	H3, H4	H1	H2	H3, H4
G 4	≥ 400	—	Determined by the insulation distance between poles AB, and C (can be mechanically realized up to min. 840 mm)	> 2254	—	—
G 9, G 10	≥ 400	≥ 400		> 2309	> 2259	—
G 11, G 12	—	≥ 400		—	> 2254	> 2249
G 13, G 14	≥ 520	≥ 400		—	> 2259	> 2259

Attention: H1 and H2 can be mechanically realized as minimum dimension, larger when the upper gear unit is turned into a certain position.

